

Labor Force Participation and the Organization of Global Value Chains*

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Abstract

This paper examines how labor supply shapes the organization of global value chains. We develop a general equilibrium model in which labor-force participation affects firms' sourcing decisions through equilibrium wages and intermediate-input costs. The model predicts that higher labor-force participation reduces backward GVC participation, increases forward GVC participation, shifts production toward upstream activities, and shortens internationally fragmented GVC production chains. Using a country-sector panel covering 53 economies and 35 sectors over 2007–2021, we estimate these relationships using an instrumental-variable strategy based on extreme temperature shocks. A one-percentage-point increase in the labor-force participation rate reduces backward GVC participation by 1.43 percentage points, increases forward GVC participation by 2.54 percentage points, and reduces average production length by 2.91 units. The effects are stronger in labor-intensive sectors and are attenuated by deeper regional trade agreements, which proxy for lower trade costs. The findings identify labor supply as an important determinant of the direction and vertical organization of global production.

JEL classification: F14, F16, J21, L23

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1 Introduction

The organization of global value chains (GVCs) shapes how economies specialize, derive gains from trade, and position themselves in international production networks. A country's role in a GVC depends not only on whether it participates in cross-border production, but also on how production stages are allocated across countries, whether it occupies an upstream or downstream position, and the extent to which production is fragmented internationally. Existing research has emphasized trade costs, technology, comparative advantage, and contractual frictions as key determinants of GVC organization. By contrast, the role of labor supply in shaping the cost and organization of production has received considerably less attention.

This question has become increasingly important. The COVID-19 pandemic, geopolitical tensions, climate change, and population aging have altered the availability and stability of labor across countries. These developments raise a broader question: does labor supply affect only the scale of production and trade, or does it also reshape the organization of global production networks? In particular, can persistent cross-country differences in labor-force participation determine whether firms source inputs domestically or fragment production across borders? Despite its importance, the relationship between labor supply and the architecture of GVCs remains insufficiently understood.

This paper argues that labor supply affects not only production capacity but also the domestic–foreign boundary and the vertical structure of global production. We incorporate labor supply into the endogenous multi-stage production framework of Antràs (2023). This framework allows us to trace how labor supply affects equilibrium wages, input costs, firms' sourcing decisions, and the number of production stages. The central mechanism is straightforward. A higher labor-force participation rate (LFPR) expands the domestic supply of labor and lowers the equilibrium wage. As a result, domestic labor-intensive intermediates become relatively cheaper, inducing firms to substitute domestic sourcing

for imported inputs. This reallocation increases the share of production undertaken domestically and shifts the country toward more upstream activities. At the same time, it reduces the optimal number of internationally fragmented production stages.

The model therefore generates predictions that go beyond the conventional effect of labor supply on trade volumes. A higher domestic LFPR reduces backward GVC participation, increases forward GVC participation, shifts a country toward a more upstream position in production networks, and shortens internationally fragmented production chains. These effects are stronger when trade costs are higher. When cross-border production sharing is more costly, an increase in domestic labor supply creates a stronger incentive to consolidate production at home. Conversely, trade integration facilitates international fragmentation and attenuates the domestic-sourcing response to greater labor availability.

We test these predictions using a country–sector panel covering 53 economies and 35 sectors over the period 2007–2021. The data are constructed from the UIBE GVC Database (Wang et al., 2013, 2017). To address the potential endogeneity of LFPR, we employ a two-stage least-squares (2SLS) strategy that instruments labor-force participation with compound extreme-climate events, measured by the mean number of hot-dry days. The relevance of this instrument is motivated by evidence that heat and drought stress reduce labor productivity, increase morbidity, and lower effective labor supply (Dell et al., 2012; Burke et al., 2015).

The empirical results are consistent with the model’s predictions. A one-percentage-point increase in domestic LFPR reduces backward GVC participation by 1.43 percentage points and increases forward GVC participation by 2.54 percentage points. It also shifts countries toward more upstream positions in production networks. These effects are approximately twice as large in labor-intensive industries, consistent with the proposed labor-cost mechanism. Higher LFPR further reduces production linkage yield (PLY) by 5.50 units and average production length (APL) by 2.91 units, indicating a less interna-

tionally fragmented and more domestically integrated production structure. Finally, trade integration moderates these effects. Deeper regional trade agreements (RTAs), which we use as a proxy for lower trade costs, attenuate the relationship between LFPR and GVC outcomes. This finding is consistent with the model’s prediction that lower trade barriers facilitate cross-border fragmentation and weaken the domestic-sourcing incentives associated with greater labor availability.

This paper makes two contributions. First, it contributes to the literature on labor supply and international trade. Existing studies have examined how outsourcing affects wage inequality (Feenstra and Hanson, 1999), how labor-market frictions shape trade and unemployment (Helpman and Itskhoki, 2010), and how trade shocks, immigration, and temporary labor-supply disruptions affect labor-market outcomes and production (Glitz, 2012; Caliendo et al., 2019; Ksoll et al., 2023; Bartik et al., 2020; del Rio-Chanona et al., 2020). This literature has largely focused on labor-market outcomes or treated labor-supply variation as a market friction or temporary shock. We instead examine how persistent differences in labor-force participation affect the organization of international production. We show that labor supply influences not only the volume of trade, but also the direction of specialization, the allocation of production stages, and the length of international production chains.

Second, we contribute to the literature on the determinants of GVC organization and production fragmentation. Previous studies have emphasized comparative advantage (Costinot et al., 2013), contractual frictions (Antràs and Chor, 2013), and trade costs (Baldwin and Venables, 2013). Related work has incorporated labor-market interactions into firm-to-firm trade linkages (Eaton et al., 2022) and examined the spatial organization of GVCs (Antràs and De Gortari, 2020). We identify labor-force participation as a distinct and underexplored determinant of GVC organization. By allowing labor supply to affect equilibrium wages and domestic input costs, our framework shows how labor-market con-

ditions shape GVC participation, the upstreamness of production, and production-chain length. It also highlights how trade costs mediate these effects: lower trade barriers facilitate international fragmentation and weaken the incentive for firms to substitute domestic for foreign sourcing. In this way, the paper extends the endogenous multi-stage production framework of Antràs (2023) by introducing labor supply as a determinant of both sourcing patterns and the optimal length of production chains.

The remainder of the paper proceeds as follows. Section 2 develops the theoretical model and derives the main testable predictions. Section 3 describes the data, identification strategy, and baseline empirical results. Section 4 examines sectoral heterogeneity. Section 5 reports robustness checks. Section 6 concludes.

2 Model and Theoretical Propositions

We develop a two-country general equilibrium trade model to study how labor supply shapes the organization and vertical structure of global production networks. Specifically, we introduce the labor-force participation rate as an exogenous labor-supply shifter into the endogenous multi-stage production structure developed by Antràs (2023). Following the logic of Antràs (2023), production unfolds through a sequential process, while the optimal number of production stages is determined endogenously by the trade-off between specialization gains and coordination costs.

2.1 Setup

There are two countries, indexed by $i \in \{h, f\}$ (h = home, f = foreign). Each country is populated by a continuum of identical households. Production takes place in two sectors: an intermediate-goods sector organized as an endogenous multi-stage chain, and a final-goods sector that assembles the terminal outputs of the two national chains into a consumption good. All markets are perfectly competitive and all technologies exhibit constant returns to scale, so that every firm earns zero profits.

Two conventions are fixed here and maintained throughout. First, an internationally traded coordination service (management, quality control, supply-chain monitoring) is supplied competitively on the world market at a constant price and serves as the numeraire,

$$p^M \equiv 1. \quad (1)$$

Home labor is not a suitable numeraire because wages are endogenous. Second, cross-border shipments of goods incur iceberg trade costs $\tau_i - 1 \geq 0$ on imports into country i ; coordination services are not subject to trade costs. Section 2.1 characterizes technologies, preferences and individual optimization; market-clearing conditions and the definition of equilibrium are collected in Section 2.2.

Assumption 1 (Coordination services). The coordination input required at each stage is purchased on the world market at the numeraire price (1). It is produced without domestic labor, and neither country holds an endowment of it, so all coordination inputs are imported services.

2.1.1 Households and Final-Good Demand

There are two countries, indexed by h (home) and f (foreign). The representative household in country h consumes two final varieties, the domestically produced variety in quantity Q_{hh} and the imported variety in quantity Q_{hf} , where the first subscript denotes the country of consumption and the second the country of production. Preferences are of the constant-elasticity-of-substitution (CES) form

$$U_h = (Q_{hh}^\gamma + Q_{hf}^\gamma)^{1/\gamma}, \quad 0 < \gamma < 1, \quad \sigma \equiv \frac{1}{1-\gamma} > 1, \quad (2)$$

where U_h is the utility of the country- h household, γ governs the curvature of preferences over the two varieties, and σ is the implied elasticity of substitution between them; $\gamma \in (0, 1)$ is equivalent to $\sigma > 1$, so the two final varieties are gross substitutes.

Let p_h and p_f denote the producer (f.o.b.) prices of the final varieties produced in h and f , respectively, and let $\tau_h \geq 1$ be the iceberg trade cost incurred on shipments from f to h , so that τ_h units must be shipped for one unit to arrive and $\tau_h p_f$ is the delivered (c.i.f.) price faced by consumers in h . The budget constraint is

$$p_h Q_{hh} + \tau_h p_f Q_{hf} = I_h, \quad (3)$$

where I_h is household income in country h . Because all producers operate under perfect competition and constant returns to scale, they earn zero profits in equilibrium; moreover, by Assumption 1 below, households hold no endowment of coordination services and hence receive no income from that factor. Household income therefore consists exclusively of labor income,

$$I_h = \omega_h E_h, \quad (4)$$

where ω_h is the wage rate in country h and E_h is aggregate employment, defined in (7) below.¹ Maximizing (2) subject to (3) yields the CES demand system

$$Q_{hh} = \left(\frac{p_h}{P_{Uh}} \right)^{-\sigma} \frac{\omega_h E_h}{P_{Uh}}, \quad Q_{hf} = \left(\frac{\tau_h p_f}{P_{Uh}} \right)^{-\sigma} \frac{\omega_h E_h}{P_{Uh}}, \quad (5)$$

where P_{Uh} is the CES consumer price index, that is, the minimum expenditure required to attain one unit of U_h , and s_{hh} is the share of country- h income spent on the domestically produced variety:

$$P_{Uh} = \left(p_h^{1-\sigma} + (\tau_h p_f)^{1-\sigma} \right)^{\frac{1}{1-\sigma}}, \quad s_{hh} \equiv \frac{p_h Q_{hh}}{\omega_h E_h} = \frac{p_h^{1-\sigma}}{p_h^{1-\sigma} + (\tau_h p_f)^{1-\sigma}}. \quad (6)$$

Real income, $W_h = \omega_h E_h / P_{Uh}$, coincides with indirect utility and serves as our welfare measure. Symmetric expressions hold for country f , with (τ_h, p_h, p_f) replaced by (τ_f, p_f, p_h) , where $\tau_f \geq 1$ is the iceberg cost of shipping from h to f .

¹Since τ_h is an iceberg trade cost rather than an ad valorem tariff, no tariff revenue accrues to households. With constant returns to scale and perfect competition, value added equals the wage bill, so $\omega_h E_h$ exhausts household income.

2.1.2 Labor Market

Consider country $i \in \{h, f\}$. The country is endowed with an exogenous population of size L_i , of which an exogenous fraction $N_i \in (0, 1]$, the labor-force participation rate, is available for work; the measure of labor-market participants is therefore $N_i L_i$. Not all participants are employed: the employment rate among participants, $H_i \in (0, 1]$, is determined endogenously in equilibrium, so that $1 - H_i$ is the unemployment rate. Aggregate employment is

$$E_i = H_i N_i L_i. \quad (7)$$

Following Helpman and Itskhoki (2010), we do not model the matching process explicitly and instead summarize wage determination by a wage-setting curve that is increasing in labor-market tightness, which, with H_i measured as the employment rate among participants, is itself increasing in H_i :

$$\omega_i = a_i H_i^{\beta_i}, \quad a_i > 0, \beta_i > 0, H_i \in (0, 1], \quad (8)$$

where the wage rate ω_i is the same object that appears in the household budget constraint (4), the shift parameter $a_i > 0$ captures country-specific labor productivity and labor-market institutions, and $\beta_i > 0$ is the elasticity of the wage with respect to tightness. A larger β_i means that a given expansion of labor demand is absorbed relatively more by wage increases and relatively less by employment gains, so that β_i governs how the gains from trade are split between the wage and the employment margin.

2.1.3 Multi-Stage Intermediate Production

The intermediate sector in country i operates a chain of $\bar{n}_i$ sequential production stages, indexed by $n = 1, \dots, \bar{n}_i$, where stage n uses the output of stage $n - 1$ as its sole intermediate input and $\bar{n}_i$, the length of the chain, is chosen optimally in Section 2.1.4. We first state the technological assumptions that govern the trade-off between the benefits and the costs of fragmenting production into a larger number of stages.

Assumption 2 (Specialization gains and numeraire-denominated coordination costs). For $i \in \{h, f\}$ and every stage $n \geq 1$:

- (a) stage productivity $A_{i,n}$, defined as output per unit of the stage- n input bundle, exhibits concave specialization gains,

$$A_{i,n} = n^c, \quad c \in (0, 1), \quad (9)$$

where c is the elasticity of stage productivity with respect to chain position, so that $A_{i,1} = 1$ and $\ln A_{i,n}$ is strictly increasing and concave in n ;

- (b) coordination is a real resource cost that accumulates additively in the number of stages: producing one unit of stage- n output requires $\chi(n-1)$ units of the coordination service, where $\chi > 0$ is the coordination requirement per additional stage, priced in the numeraire defined in (1);
- (c) the primary stage uses labor only, so its labor share satisfies $\lambda_{i,1} = 1$, and it requires no coordination.

Part (a) states that productivity rises with the position of a stage in the chain, while the marginal gain from further fragmentation diminishes because $c < 1$.² Part (b) implies that fragmentation economizes on a wage-intensive cost component while incurring an overhead whose price is pinned down outside the domestic labor market, and part (c) anchors the chain in a stage whose cost is purely a labor cost. Taken together, the three parts deliver an interior optimal chain length: the benefit of adding a stage is concave in n , whereas its coordination cost is linear in n .

²The parameter c measures the attenuation of marginal specialization gains: as c approaches one, the gains decay slowly, reflecting strong division-of-labor synergies; as c approaches zero, they are exhausted rapidly.

Stage 1 (the primary intermediate). Firms at the first stage use domestic labor as their only input:

$$X_{i,1} = A_{i,1} l_{i,1}, \quad P_{i,1} = \frac{\omega_i}{A_{i,1}} = \omega_i, \quad (10)$$

where $X_{i,1}$ is the quantity of stage-1 output produced in country i , $l_{i,1}$ is the labor input employed at that stage, and $P_{i,1}$ is the associated unit cost, which under perfect competition and constant returns to scale equals the producer price. The second equality follows from $A_{i,1} = 1^c = 1$ in Assumption 2(a), and stage 1 carries no coordination overhead by Assumption 2(c); the primary intermediate is therefore priced at the domestic wage.

Stages $n \geq 2$: recursive production with cross-border sourcing. At each subsequent stage $n \in \{2, \dots, \bar{n}_i\}$, a competitive firm combines the stage- $(n-1)$ input, procured from the cheaper of the two sources and available in quantity $\hat{X}_{i,n-1}$, with domestic labor $l_{i,n}$ according to the Cobb-Douglas technology

$$X_{i,n} = A_{i,n} (\hat{X}_{i,n-1})^{1-\lambda_{i,n}} l_{i,n}^{\lambda_{i,n}}, \quad \lambda_{i,n} \in (0, 1), \quad (11)$$

where $\lambda_{i,n}$ is the labor share in the stage- n cost, so that $1 - \lambda_{i,n}$ is the cost share of the upstream input; this is the sense in which Assumption 2(c) sets $\lambda_{i,1} = 1$. The upstream input is bought at the cost-minimizing delivered price

$$\hat{P}_{i,n-1} = \min\{P_{i,n-1}, \tilde{\tau}_i P_{j,n-1}\}, \quad \tilde{\tau}_i P_{j,n-1} \equiv \begin{cases} \tau_i P_{j,n-1}, & n-1 \leq \bar{n}_j, \\ +\infty, & n-1 > \bar{n}_j, \end{cases} \quad j \neq i, \quad (12)$$

where $P_{i,n-1}$ is the domestic unit cost of stage $n-1$, $P_{j,n-1}$ is its unit cost in the partner country $j \neq i$, and $\tau_i \geq 1$ is the iceberg cost of shipping intermediates from j to i , so that $\tau_i P_{j,n-1}$ is the delivered price of the imported input. The convention embedded in (12) makes explicit that a stage can be imported only if the partner country actually operates it, and ties are broken in favor of domestic sourcing. Accordingly, the set of stages that country i imports rather than performs at home is

$$\mathcal{K}_i \equiv \{n \in \{1, \dots, \bar{n}_i - 1\} : n \leq \bar{n}_j \text{ and } \tau_i P_{j,n} < P_{i,n}\}, \quad j \neq i. \quad (13)$$

Because $\mathcal{K}_i$ depends on the foreign cost path $\{P_{j,n}\}_{n=1}^{\bar{n}_j}$, which in turn depends on the foreign sourcing set $\mathcal{K}_j$ and the foreign chain length $\bar{n}_j$, the two sourcing patterns and the two chain lengths are determined jointly in equilibrium.

Stage prices. Under perfect competition and constant returns to scale, the user price of stage- n output equals its variable unit cost plus the coordination overhead introduced in Assumption 2(b). Substituting the productivity schedule (9) into the unit-cost function implied by the technology (11) gives

$$P_{i,1} = \omega_i, \quad \underbrace{P_{i,n}}_{\text{user price}} = \underbrace{\frac{1}{n^c} \left(\frac{\hat{P}_{i,n-1}}{1 - \lambda_{i,n}} \right)^{1 - \lambda_{i,n}} \left(\frac{\omega_i}{\lambda_{i,n}} \right)^{\lambda_{i,n}}}_{\equiv V_{i,n} \text{ (variable cost)}} + \underbrace{\chi(n-1)}_{\text{coordination overhead}}, \quad n \geq 2, \quad (14)$$

where $V_{i,n}$ is the variable unit cost of stage n in country i , that is, the minimum outlay on the upstream input and on labor required to produce one unit of stage- n output, given the delivered input price $\hat{P}_{i,n-1}$ from (12), the wage ω_i , and the labor share $\lambda_{i,n}$. Two properties of (14) drive the results below. First, $V_{i,n}$ is homogeneous of degree one in $(\hat{P}_{i,n-1}, \omega_i)$ and is scaled down by the specialization factor n^{-c} , so lengthening the chain reduces variable cost at a decreasing rate. Second, by Assumption 1 the overhead $\chi(n-1)$ is a real resource cost denominated in the numeraire and is therefore independent of the domestic wage. Fragmentation consequently substitutes a wage-independent overhead for a wage-intensive cost component, which is the mechanism through which chain length interacts with the labor market.

Cost-minimizing factor allocation. Cost minimization of the Cobb–Douglas bundle in (11) implies that expenditure on each input is a constant share of variable cost, so that

$$\omega_i l_{i,n}^* = \lambda_{i,n} V_{i,n} X_{i,n}, \quad \hat{P}_{i,n-1} \hat{X}_{i,n-1}^* = (1 - \lambda_{i,n}) V_{i,n} X_{i,n}, \quad M_{i,n}^{\text{coord}} = \chi(n-1) X_{i,n}, \quad (15)$$

where $X_{i,n}$ denotes total stage- n output of country i , that is, the sum of quantities absorbed domestically and exported; $l_{i,n}^*$ is the cost-minimizing quantity of labor hired by country i stage- n producers; $\hat{X}_{i,n-1}^*$ is their cost-minimizing demand for the stage- $(n-1)$ input, measured at delivery and purchased at the price $\hat{P}_{i,n-1}$ defined in (12); and $M_{i,n}^{\text{coord}}$ is the quantity of the imported coordination service they require, which is proportional to output by Assumption 2(b) and valued at the numeraire price (1). Summing $M_{i,n}^{\text{coord}}$ across stages $n \geq 2$ delivers country i 's aggregate coordination imports D_i^M , which enter the trade-balance condition in Section 2.2. The first two expressions in (15) state that $\lambda_{i,n}$ and $1 - \lambda_{i,n}$ are the labor and upstream-input shares of variable cost; the third states that coordination requirements scale linearly with output and, unlike the first two, do not respond to relative prices. Because prices equal unit costs, revenue is exhausted stage by stage, $P_{i,n}X_{i,n} = V_{i,n}X_{i,n} + \chi(n-1)X_{i,n}$, so producers earn zero profits and the entire value of output accrues to labor, to upstream suppliers, and to the coordination input. Combining this accounting identity with (15) gives stage-level labor demand,

$$l_{i,1}^* = \frac{P_{i,1}X_{i,1}}{\omega_i} = X_{i,1}, \quad l_{i,n}^* = \frac{\lambda_{i,n}V_{i,n}X_{i,n}}{\omega_i}, \quad n \geq 2. \quad (16)$$

Stage 1 uses labor alone, so its labor requirement is simply its output; at later stages labor demand falls in the wage, both directly and indirectly through $V_{i,n}$. Aggregation of (16) into economy-wide labor demand is deferred to the labor-market-clearing condition in Section 2.2.

2.1.4 Optimal Length of the Production Chain

Because every stage operates under perfect competition and earns zero profits, minimizing the delivered price of the terminal intermediate good is equivalent to cost minimization for any agent along the chain; it is therefore immaterial which party is assigned the choice of chain length, and we let the downstream final-goods firm select the number of stages

$$\bar{n}_i = \arg \min_{n \in \mathbb{N}^+} P_{i,n}, \quad i \in \{h, f\}, \quad (17)$$

where $\bar{n}_i$, the equilibrium length of country i 's production chain, denotes the number of sequential stages actually operated. Existence of a solution follows from (14): since $P_{i,n} = V_{i,n} + \chi(n-1)$ with $V_{i,n} \geq 0$ and $P_{i,1} = \omega_i < \infty$, the objective in (17) diverges as n grows without bound whenever $\chi > 0$, so the minimum is attained at a finite $\bar{n}_i$. It is convenient to work with the incremental cost of operating one additional stage,

$$\Delta_i(n; \omega_i, \tau_i) \equiv P_{i,n+1} - P_{i,n} = [V_{i,n+1} - V_{i,n}] + \chi, \quad (18)$$

where $\Delta_i(\cdot)$ measures the change in the delivered price when the chain is lengthened from n stages to $n+1$ stages, holding the wage ω_i and the trade cost τ_i fixed; we adopt the convention $\Delta_i(0) \equiv -\infty$, which guarantees that the first stage is always operated. Any solution to (17) must satisfy the economic requirement that the last stage adopted is weakly worthwhile and the next stage is weakly unprofitable,

$$\Delta_i(\bar{n}_i - 1; \omega_i, \tau_i) \leq 0 \leq \Delta_i(\bar{n}_i; \omega_i, \tau_i); \quad (19)$$

in particular, $\bar{n}_i = 1$ implies $\Delta_i(1) \geq 0$, and the converse holds under Assumption 3 below. This assumption rules out multiple crossings of $\Delta_i(\cdot)$ and hence renders (19) sufficient as well as necessary for optimality. Two opposing forces determine the sign of $\Delta_i(n)$. The first is specialization gains. Because $A_{i,n} = n^c$ is increasing in n , a longer chain lowers variable unit cost $V_{i,n}$ for a given input price and thus tends to make $V_{i,n+1} - V_{i,n}$ negative. The magnitude of this saving is governed by the wage. When $\chi = 0$ and the chain is sourced entirely at home ($\mathcal{K}_i = \emptyset$), every stage price is homogeneous of degree one in ω_i , so the saving scales one for one with the wage. When $\chi > 0$ this exact homogeneity fails, because the input price $P_{i,n-1} = V_{i,n-1} + \chi(n-2)$ contains a wage-invariant component; define the wage elasticity of variable cost as $\varepsilon_{i,n}^V \equiv \partial \ln V_{i,n} / \partial \ln \omega_i$, which lies in $(0, 1]$, with strict inequality for $n \geq 3$. The second countervailing force is the coordination overhead χ incurred for each additional stage, which by Assumption 1, is invariant to ω_i . A lower wage therefore shrinks the variable-cost saving relative to this fixed overhead, and, under

the conditions stated below, shortens the chain. This asymmetry between a wage-elastic benefit and a wage-invariant cost of fragmentation constitutes the mechanism through which labor-market outcomes feed back into the organization of production.

The two forces in $\Delta_i(n)$ can be made explicit. Decompose each stage price into its wage-proportional and wage-invariant parts,

$$P_{h,n} = \omega_h v_{h,n} + \chi(n-1), \quad (20)$$

where $v_{h,n}$ is the unit variable labor requirement of stage n , defined for given delivered upstream prices $\{\hat{P}_{h,n-1}\}_n$; when $\mathcal{K}_h = \emptyset$ the decomposition is exact and $P_{h,n}$ is homogeneous of degree one in ω_h net of the overhead. The following lemma shows that the wage-proportional component declines with the number of stages, so that lengthening the chain buys a saving proportional to ω_h at a cost proportional to χ .

Lemma 1 (Declining unit variable labor requirement). *Under Assumptions 2, the unit variable labor requirements defined by (20) satisfy*

$$v_{h,1} = 1 \quad \text{and} \quad v_{h,n+1} < v_{h,n} \leq 1 \quad \text{for every } n \geq 1.$$

Proof. Substituting (20) into the price recursion (14) and collecting the terms proportional to ω_h gives

$$v_{h,n+1} = A_{h,n+1}^{-1} \left[\lambda_{h,n+1} + (1 - \lambda_{h,n+1}) v_{h,n} \right], \quad (21)$$

a convex combination of 1 and $v_{h,n}$ scaled by $A_{h,n+1}^{-1}$. We prove the claim by induction. For the base step, $P_{h,1} = \omega_h$ gives $v_{h,1} = 1$, and (21) then yields $v_{h,2} = 2^{-c} < 1 = v_{h,1}$. For the inductive step, suppose $v_{h,n} < v_{h,n-1} \leq 1$. Since $\lambda + (1 - \lambda)v$ is increasing in λ whenever $v < 1$, and $\lambda_{h,n+1} \leq \lambda_{h,n}$, the bracket in (21) is weakly smaller than its stage- n counterpart; and since $A_{h,n+1} = (n+1)^c > n^c = A_{h,n}$ for $c \in (0, 1)$ by Assumption 2, the scaling factor is strictly smaller. Hence $v_{h,n+1} < v_{h,n}$, and $v_{h,n+1} < 1$ because the bracket is at most one. The overhead $\chi(n-1)$ is wage-invariant and does not enter (21), which is

precisely why fragmentation trades a wage-proportional saving against a wage-invariant cost. When $\mathcal{K}_h \neq \emptyset$ the same recursion applies with $\hat{P}_{h,n-1}$ in place of $P_{h,n-1}$, so a higher τ_h raises $v_{h,n}$ at every stage that would otherwise be imported. $\square$

By (20) and Lemma 1, the marginal gain from an additional stage satisfies

$$\frac{\partial \Delta_h(n)}{\partial \omega_h} = -(v_{h,n} - v_{h,n+1}) < 0,$$

so a lower wage raises the marginal gain from fragmentation at every n . This is the property that Assumption 3 extends to the full organizational problem.

Assumption 3 (Strict single crossing of the stage margin). Let $\Delta_i(n; \omega_i, \tau_i) \equiv P_{i,n+1} - P_{i,n}$ be the marginal stage margin defined in (18), that is, the change in the user price of the final good in country $i \in \{h, f\}$ when the chain is lengthened from n to $n + 1$ stages, evaluated at the wage $\omega_i > 0$ and the trade cost τ_i . For each $i \in \{h, f\}$ and every $\omega_i > 0$, the function $\Delta_i(\cdot; \omega_i, \tau_i)$ crosses zero exactly once and from below in n : there exists an integer $n^* = n^*(\omega_i, \tau_i)$ such that $\Delta_i(n; \omega_i, \tau_i) < 0$ for all $n < n^*$ and $\Delta_i(n; \omega_i, \tau_i) > 0$ for all $n \geq n^*$. Economically, the specialization gain from a finer division of labor dominates the additional coordination overhead at short chains, whereas the overhead dominates once the chain is sufficiently long, and the ranking switches only once. Consequently, the stopping condition (19) selects the unique global minimizer $\bar{n}_i = n^*(\omega_i, \tau_i)$ of the price objective (17), so that the optimal chain length is well defined and no local comparison of adjacent stages can be mistaken for the global optimum.

2.1.5 Aggregation of Domestic and Imported Intermediates

The composite intermediate used by the final-goods firm in country h is a CES aggregate of the terminal outputs of the two national chains. The relevant prices are the terminal stage prices generated by (14), namely $P_{h,\bar{n}_h}$ for the domestic variety and $P_{f,\bar{n}_f}$ for the imported variety, where $\bar{n}_i$ is the optimal chain length of country i characterized in (19).

Hence the endogenous chain lengths and sourcing patterns enter the downstream block only through these two prices. The composite is

$$X_h = \left(X_{hh}^{\frac{\sigma_{Xh}-1}{\sigma_{Xh}}} + X_{hf}^{\frac{\sigma_{Xh}-1}{\sigma_{Xh}}} \right)^{\frac{\sigma_{Xh}}{\sigma_{Xh}-1}}, \quad (22)$$

where X_h is the composite intermediate quantity, X_{hh} is the quantity of the domestically produced variety, X_{hf} is the quantity of the imported variety, and $\sigma_{Xh} \equiv 1 + \alpha_h > 1$ is the elasticity of substitution between the two varieties, with $\alpha_h > 0$ the preference or technology parameter governing their substitutability. Cost minimization of the expenditure needed to attain a given X_h yields the associated price index and the conditional input demands

$$P_{Xh} = \left(P_{h,\bar{n}_h}^{1-\sigma_{Xh}} + (\tau_h P_{f,\bar{n}_f})^{1-\sigma_{Xh}} \right)^{\frac{1}{1-\sigma_{Xh}}}, \quad (23)$$

$$X_{hh} = \left(\frac{P_{h,\bar{n}_h}}{P_{Xh}} \right)^{-\sigma_{Xh}} X_h, \quad X_{hf} = \left(\frac{\tau_h P_{f,\bar{n}_f}}{P_{Xh}} \right)^{-\sigma_{Xh}} X_h, \quad (24)$$

where P_{Xh} is the unit cost of the composite intermediate, that is, the minimized expenditure per unit of X_h , and $\tau_h \geq 1$ is the iceberg trade cost of shipping the foreign terminal variety to country h . Two features of (23) and (24) are worth emphasizing. First, the imported variety is valued at its delivered price $\tau_h P_{f,\bar{n}_f}$, so trade costs shape sourcing at two distinct margins: within the chain, through the delivered price of offshored stage inputs in (12), and across chains, through the terminal-variety price index in (23). Second, since $\sigma_{Xh} = 1 + \alpha_h > 1$, the two varieties are gross substitutes, so a rise in the relative price of one variety lowers both its relative quantity and its share in intermediate expenditure. Analogous expressions hold for country f , with $\sigma_{Xf} \equiv 1 + \alpha_f$ replacing σ_{Xh} and τ_f replacing τ_h .

2.1.6 Final-Goods Production

The final-goods market in each country is perfectly competitive. A representative firm in country h combines labor with the composite intermediate defined in (22) according to

the constant-returns-to-scale technology

$$F_h = (l_h^{fin})^{\lambda^{fin}} X_h^{1-\lambda^{fin}}, \quad \lambda^{fin} \in (0, 1), \quad (25)$$

where F_h is final-goods output, l_h^{fin} is labor employed in final-goods production, X_h is the composite intermediate from (22), λ^{fin} is the cost share of labor in final-goods production, and total factor productivity is normalized to unity. Because (25) is Cobb–Douglas and hence homogeneous of degree one, free entry and zero profits imply that the producer price of the final good equals its unit cost,

$$p_h = \left(\frac{P_{Xh}}{1 - \lambda^{fin}} \right)^{1-\lambda^{fin}} \left(\frac{\omega_h}{\lambda^{fin}} \right)^{\lambda^{fin}}, \quad (26)$$

where p_h is the price of the final good in country h , P_{Xh} is the composite-intermediate price index from (23), and ω_h is the wage in country h . Applying Shephard’s lemma to (26) delivers the conditional factor demands in value form,

$$\omega_h l_h^{fin} = \lambda^{fin} p_h F_h, \quad P_{Xh} X_h = (1 - \lambda^{fin}) p_h F_h. \quad (27)$$

Thus expenditure on labor and on the composite intermediate absorbs constant shares λ^{fin} and $1 - \lambda^{fin}$ of final-goods revenue, and the entire influence of chain length, offshoring and trade costs on the downstream block operates through P_{Xh} in (26). Equations (25)–(27) complete the description of individual optimization. All market-clearing conditions, namely those for final goods, stage-level intermediates, coordination services and labor, together with the balance of payments, the full list of endogenous variables and the definition of equilibrium, are stated in Section 2.2.

2.2 Market Clearing and Equilibrium

Section 2.1 described preferences, technologies and individual optimization. This subsection closes the model: it collects all market-clearing conditions, defines competitive equilibrium, and establishes existence and generic local uniqueness. Throughout, all prices are

expressed in units of the numeraire, the internationally traded coordination service, whose world price is normalized to $p^M \equiv 1$ in (1).

2.2.1 Market Clearing

Coordination-service market. By Assumption 2, producing one unit of stage- n output requires $\chi(n-1)$ units of the coordination service, where $\chi > 0$ is the per-stage coordination overhead and $n-1$ counts the interfaces between the n stages of the chain. Aggregating this requirement over all stages operated by country i , the country's flow demand for coordination services is

$$D_i^M = \chi \sum_{n=2}^{\bar{n}_i} (n-1) X_{i,n}, \quad (28)$$

where D_i^M is country i 's total imports of the coordination service, $X_{i,n}$ denotes gross stage- n output including quantities produced for export, and $\bar{n}_i$ is the optimal chain length determined by (19). The summation begins at $n=2$ because the initial primary stage involves no inter-stage interface and hence incurs no coordination requirement. Under Assumption 1, world supply of the coordination service is perfectly elastic at $p^M = 1$; therefore (28) pins down the quantity D_i^M absorbed by country i at that price, and no separate world-clearing condition for this market needs to be imposed.

Final-goods markets. Final-goods markets clear country by country,

$$F_h = Q_{hh} + Q_{fh}, \quad F_f = Q_{ff} + Q_{hf}, \quad (29)$$

where F_i denotes final-goods output of country i , and Q_{ij} is the quantity of country- i final good delivered to consumers in country j , with demands given by (5) and its foreign analogue. All quantities are measured at delivery, so that the volume shipped from i to $j \neq i$ exceeds the delivered volume by the iceberg factor τ_j .

Stage-level intermediate-goods markets. For $j \neq i$, let $M_{j,n}$ denote the quantity of stage- n output produced in country i and purchased by country j for use by its stage- $(n+1)$ producers, and let X_{ji} denote the quantity of country- i terminal output purchased by country j for its composite aggregator. Let $\mathcal{K}_j$ be the set of stages that country j offshores, defined in (13), and let $\mathbf{1}[\cdot]$ be the indicator function. For each active stage $n = 1, \dots, \bar{n}_i$, gross stage output is absorbed by three channels,

$$X_{i,n} = X_{i,n}^{\text{dom}} + \mathbf{1}[n \in \mathcal{K}_j] M_{j,n} + \mathbf{1}[n = \bar{n}_i] X_{ji}, \quad j \neq i, \quad (30)$$

where $X_{i,n}^{\text{dom}}$ is domestic absorption, taken up by country- i stage- $(n+1)$ producers when $n < \bar{n}_i$ and by country- i composite aggregator when $n = \bar{n}_i$. The corresponding downstream input demand follows from the cost shares in (15) together with the delivered input price $\hat{P}_{i,n-1}$ in (12):

$$\begin{aligned} \hat{X}_{i,n-1}^* &= \frac{(1 - \lambda_{i,n}) V_{i,n} X_{i,n}}{\hat{P}_{i,n-1}}, & X_{i,n-1}^{\text{dom}} &= \mathbf{1}[n-1 \notin \mathcal{K}_i] \hat{X}_{i,n-1}^*, \\ M_{i,n-1} &= \mathbf{1}[n-1 \in \mathcal{K}_i] \tau_i \hat{X}_{i,n-1}^*, \end{aligned} \quad (31)$$

where $\hat{X}_{i,n-1}^*$ is the cost-minimizing quantity of the stage- $(n-1)$ input measured at delivery, $\lambda_{i,n}$ is the labor cost share of stage n , and $V_{i,n}$ is its variable unit cost; multiplication by τ_i converts delivered into shipped quantities whenever the input crosses a border.

Two features of (30) deserve emphasis. First, trade in stage n is governed by the sourcing set $\mathcal{K}_j$ rather than by a comparison of chain lengths: operating a positive number of downstream stages abroad is necessary but not sufficient for stage n to be traded. Second, the two indicators in (30) never apply simultaneously, as established by the following lemma.

Lemma 2 (No two-way trade at a given stage). *Let $\mathcal{K}_i$ be the set of stages offshored by country i , as defined in (13), and let $\tau_h, \tau_f \geq 1$ be the iceberg trade costs. Then for every stage n the events $n \in \mathcal{K}_h$ and $n \in \mathcal{K}_f$ are mutually exclusive. Each stage is therefore*

either sourced domestically in both countries or traded in exactly one direction, so the two indicators in (30) never take the value one at the same time.

Proof. By the definition of the sourcing set in (13), $n \in \mathcal{K}_h$ requires that the delivered foreign price of stage- n output be strictly below the domestic one, that is $\tau_h P_{f,n} < P_{h,n}$, and symmetrically $n \in \mathcal{K}_f$ requires $\tau_f P_{h,n} < P_{f,n}$. Suppose both held. Since all stage prices are strictly positive, the two inequalities may be multiplied side by side, giving $\tau_h \tau_f P_{h,n} P_{f,n} < P_{h,n} P_{f,n}$ and hence $\tau_h \tau_f < 1$, which contradicts $\tau_h, \tau_f \geq 1$. Therefore at most one of the two events occurs. $\square$

Labor market. Labor is freely mobile across the two sectors within country i , so a single wage ω_i clears the national labor market. Sectoral labor demands are obtained from the final-goods factor demand (27) and the stage-level labor demand (16). Labor is remunerated out of the *variable* unit cost $V_{i,n}$ rather than out of the user price $P_{i,n}$, because the coordination overhead $\chi(n-1)$ accrues to suppliers of coordination services. Using $V_{i,1} = P_{i,1} = \omega_i$ and $\lambda_{i,1} = 1$, market clearing for labor reads

$$\underbrace{\frac{\lambda_i^{fin} p_i F_i}{\omega_i}}_{l_i^{fin}} + \underbrace{\frac{1}{\omega_i} \left[\omega_i X_{i,1} + \sum_{n=2}^{\bar{n}_i} \lambda_{i,n} V_{i,n} X_{i,n} \right]}_{l_i^{int} = \sum_{n=1}^{\bar{n}_i} l_{i,n}^*} = E_i = H_i N_i L_i, \quad (32)$$

where l_i^{fin} and l_i^{int} denote employment in the final-goods sector and in the intermediate-goods stages, E_i is aggregate employment, L_i the working-age population, $N_i \in (0, 1]$ the participation rate, and $H_i \in (0, 1]$ the employment rate.

Conditions (8), (7) and (32) jointly pin down the triple (ω_i, H_i, E_i) for given sectoral output: (32) represents the demand side, while (8) represents wage setting. This block is where the participation rate operates. A rise in N_i enlarges the participating pool $N_i L_i$; holding labor demand fixed, (32) then requires a lower employment rate H_i , and (8) in turn delivers a lower equilibrium wage ω_i . Because $V_{i,n}$ is increasing in ω_i whereas the

coordination overhead is independent of it, a fall in ω_i compresses the specialization saving relative to the coordination penalty in (18). This is the channel through which N_i shapes the optimal chain length $\bar{n}_i$ and the set of offshored stages $\mathcal{K}_i$.

Balance-of-payments. Iceberg trade costs dissipate real resources rather than generating tariff revenue, and all firms earn zero profits; external accounts therefore balance country by country. For country i , goods exports valued f.o.b. equal goods imports valued c.i.f. plus the net coordination position, all expressed in units of the numeraire (1):

$$\underbrace{p_i X_{ji} + \sum_{n \in \mathcal{K}_j} P_{i,n} M_{j,n}}_{\text{goods exports}} = \underbrace{\tau_i p_j X_{ij} + \sum_{n \in \mathcal{K}_i} \tau_i P_{j,n} M_{i,n}}_{\text{goods imports}} + \underbrace{D_i^M}_{\text{coordination imports}}, \quad i \in \{h, f\}, j \neq i, \quad (33)$$

where X_{ij} denotes final goods produced in country j and absorbed in country i , $M_{i,n}$ denotes stage- n intermediate inputs sourced by country i from country j , and D_i^M denotes country i 's net purchases of coordination services from abroad.

The summation ranges are the offshoring sets $\mathcal{K}_i$ and $\mathcal{K}_j$, consistent with the stage level clearing condition (30). Equation (33) is equivalent to the statement that household income $I_i = \omega_i E_i + \bar{D}_i^M$ in (4) equals domestic absorption plus coordination purchases D_i^M , which verifies the claim made in Section 2.1.1. Given (28)–(32) for both countries, one of the two balance-of-payments conditions is redundant by Walras' law; the remaining condition pins down the relative wage $\omega \equiv \omega_h/\omega_f$.

2.2.2 Definition of Equilibrium

It is convenient to separate the discrete organizational margins from the continuous price quantity block. Define the organizational configuration

$$d \equiv (\bar{n}_h, \bar{n}_f, \mathcal{K}_h, \mathcal{K}_f), \quad \mathcal{K}_i \subseteq \{1, \dots, \min(\bar{n}_i - 1, \bar{n}_j)\}, \quad j \neq i, \quad (34)$$

where $\bar{n}_i$ is the chain length chosen in country i and $\mathcal{K}_i$ is the set of stages that country i offshores to country j . The upper bound $\min(\bar{n}_i - 1, \bar{n}_j)$ reflects two restrictions: the final stage $\bar{n}_i$ is never offshored, and an offshored stage must be operated abroad, so it cannot exceed the foreign chain length. Let $\mathcal{D}$ denote the set of admissible configurations. Because the variable unit cost satisfies $V_{i,n} \geq 0$ while the coordination overhead $\chi(n - 1)$ diverges as n grows, no configuration with $\bar{n}_i$ above some finite bound $\bar{N}(\chi)$ can minimize the stage price $P_{i,n}$; hence $\mathcal{D}$ is a finite set. Whenever the cost-minimizing configuration is not unique, ties are broken in favor of domestic sourcing and, among chains of equal cost, in favor of the shorter chain.

Definition 1 (Competitive equilibrium). *Fix the following primitives: labor endowments (L_h, L_f) , participation rates (N_h, N_f) , iceberg trade costs (τ_h, τ_f) , technology $(\lambda^{fin}, \{\lambda_{i,n}\}, c)$, preference parameters $(\gamma, \alpha_h, \alpha_f)$, wage-curve parameters $(a_h, a_f, \beta_h, \beta_f)$, the coordination wedge χ , and the numeraire $p^M \equiv 1$ as in (1). A competitive equilibrium consists of an organizational configuration $d^* \in \mathcal{D}$, defined in (34), together with a core vector*

$$\left\{ \omega_i, H_i, E_i, \{P_{i,n}\}_{n \leq \bar{n}_i}, P_{Xi}, P_{Ui}, l_i^{fin}, l_i^{int}, D_i^M \right\}_{i \in \{h,f\}},$$

where P_{Ui} is the consumer price index of country i , such that:

- (i) households optimize: (5)–(6) hold subject to the budget constraint (3) with income given by (4);
- (ii) final- and intermediate-goods firms minimize cost and earn zero profits: (14), (15), (23)–(24) and (26)–(27) hold;
- (iii) sourcing decisions satisfy (12)–(13), and each chain length solves (17), equivalently satisfies (19) under Assumption 3, both evaluated at equilibrium wages;
- (iv) wages satisfy the wage-setting curve (8) and employment satisfies $E_i = H_i N_i L_i$ with $H_i \in (0, 1)$, interiority being guaranteed by Lemma 4 below;

(v) *all markets clear: coordination services (28), final goods (29), stage-level intermediates (30) and labor (32); and the balance of payments (33) holds.*

Counting. Conditional on a configuration $d \in \mathcal{D}$, the continuous block consists of the stage-price recursion ($\sum_i \bar{n}_i$ equations), the two aggregator price indices, the two final-good prices, the two consumer price indices, the two wage curves, the two employment identities, the two labor-market clearing conditions, the two final-goods clearing conditions, the stage-level clearing conditions, the coordination-market conditions, and one balance-of-payments condition, the other being redundant by Walras' law. The numeraire $p^M \equiv 1$ in (1) removes the remaining degree of freedom. The system therefore determines relative prices and the relative wage $\omega = \omega_h/\omega_f$, as is standard in models without an outside nominal anchor.

We now characterize how stage prices respond to wages, which is the building block both for the labor-demand schedule and for the comparative statics of Section 2.4. Recall from (14) that the stage price decomposes as $P_{i,n} = V_{i,n} + \chi(n-1)$, the sum of the variable unit cost and the cumulated coordination overhead. Define

$$s_{i,n} \equiv \frac{V_{i,n}}{P_{i,n}} = \frac{V_{i,n}}{V_{i,n} + \chi(n-1)} \in (0, 1], \quad s_{i,1} = 1, \quad (35)$$

which is the share of variable cost in the stage- n price and therefore measures the elasticity of $P_{i,n}$ with respect to $V_{i,n}$.

Lemma 3 (Stage prices increase in wages). *Fix a configuration $d \in \mathcal{D}$. Then, for every active stage $n \leq \bar{n}_i$, the stage price $P_{i,n}$ is continuous and strictly increasing in the domestic wage ω_i , and its elasticity $\varepsilon_{i,n} \equiv \partial \ln P_{i,n} / \partial \ln \omega_i \in (0, 1]$ satisfies the recursion*

$$\varepsilon_{i,n} = s_{i,n} \left[\lambda_{i,n} + (1 - \lambda_{i,n}) \varepsilon_{i,n-1} \mathbf{1}\{n-1 \notin \mathcal{K}_i\} \right], \quad \varepsilon_{i,1} = 1, \quad (36)$$

where $s_{i,n}$ is the variable-cost share (35) and $\mathbf{1}\{\cdot\}$ is the indicator function. Moreover, $\varepsilon_{i,n} < 1$ for all $n \geq 2$ whenever $\chi > 0$: pass-through is incomplete beyond the first stage.

Consequently, the aggregator price index P_{X_i} , the final-good price p_i and the consumer price index P_{U_i} are continuous and strictly increasing in ω_i .

Proof. Monotonicity. Stack the stage prices as $P_n \equiv (P_{h,n}, P_{f,n})'$ and write the pricing equation (14) together with the sourcing rule (12) as the coupled map

$$P_n = \Phi_n(P_{n-1}; \omega_h, \omega_f), \quad P_1 = (\omega_h, \omega_f)'.$$

Each component of Φ_n is a minimum of finitely many functions that are continuous and strictly increasing in P_{n-1} and in ω_i ; hence each component inherits both properties, since continuity and strict monotonicity are preserved under the pointwise minimum of finitely many such functions. Forward induction on n therefore determines $\{P_n\}_{n \leq \max_i \bar{n}_i}$ uniquely and delivers monotonicity. When $\mathcal{K}_i \neq \emptyset$ the two national recursions must be solved jointly, because a stage $n \in \mathcal{K}_i$ prices its input at the delivered foreign price $\tau_i P_{j,n-1}$.

Elasticity. The bound follows because the coordination overhead $\chi(n-1)$ is wage invariant. Log-differentiating $P_{i,n} = V_{i,n} + \chi(n-1)$ with respect to ω_i gives $\varepsilon_{i,n} = s_{i,n} \partial \ln V_{i,n} / \partial \ln \omega_i$, with $s_{i,n}$ as in (35). By the sourcing rule (12), $\partial \ln V_{i,n} / \partial \ln \omega_i = \lambda_{i,n} + (1 - \lambda_{i,n}) \varepsilon_{i,n-1} \mathbf{1}\{n-1 \notin \mathcal{K}_i\} \in (0, 1]$, which yields (36). Since $s_{i,n} < 1$ for $n \geq 2$ when $\chi > 0$, the recursion implies $\varepsilon_{i,n} < 1$. The formal induction is deferred to Lemma 8. $\square$

Assumption 4 (Downward-sloping aggregate labor demand). For every $d \in \mathcal{D}$, the aggregate labor demand $E_i^d(\omega_i)$, defined as the left-hand side of the labor-market clearing condition (32), is continuously differentiable and strictly decreasing in ω_i , with elasticity

$$\eta_i \equiv - \frac{\partial \ln E_i^d(\omega_i)}{\partial \ln \omega_i} > 0,$$

and satisfies the boundary conditions $\lim_{\omega_i \rightarrow 0^+} E_i^d(\omega_i) = +\infty$ and $\lim_{\omega_i \rightarrow \infty} E_i^d(\omega_i) = 0$. Here $d \in \mathcal{D}$ indexes the organizational configuration (34) and ω_i denotes the relative wage consistent with the numeraire normalization (1).

Assumption 4 is imposed directly on the aggregate labor demand schedule rather than derived from the demand primitives, because monotonicity does not follow from CES preferences alone. With multi-stage input-output linkages, iceberg trade costs and endogenous sourcing sets, aggregate labor demand combines stage-level cost pass-through with sourcing substitution, so its slope is a general-equilibrium object rather than a property of the demand system. Lemma 3 supplies the two ingredients that make the assumption plausible: pass-through is bounded, since $\varepsilon_{i,n} \in (0, 1]$ by (36), and the sourcing shares respond to relative costs according to (53) with elasticity $\sigma_{Xi} = 1 + \alpha_i > 1$. Both channels push labor demand down when ω_i rises; the assumption rules out the knife-edge case in which offsetting reallocation across stages overturns this effect. We state the condition separately for two reasons. First, η_i is the only demand-side object that enters the comparative statics of Section 2.4. Second, η_i must be distinguished from the *stage-level* pass-through elasticity $\varepsilon_{i,n}$ of Lemma 3: η_i is an aggregate *quantity* elasticity with respect to the wage, whereas $\varepsilon_{i,n}$ is a stage-specific *price* elasticity.

Equilibrium wages are characterized in two steps: we first solve each country's labor market taking the other country's wage as given, and then close the two-country system.

(i) Conditional wage. Fix a configuration $d \in \mathcal{D}$ and a foreign wage ω_j , $j \neq i$. Let $E_i^d(\omega_i; \omega_j)$ denote the left-hand side of the labor-market clearing condition (32), now written explicitly as a function of both wages, and let

$$E_i^s(\omega_i) = H_i(\omega_i) N_i L_i, \quad H_i(\omega_i) = (\omega_i/a_i)^{1/\beta_i},$$

denote aggregate labor supply, where $H_i(\omega_i)$ is the employment rate implied by the wage curve (8), N_i is the labor-participation rate and L_i is the population. Define the excess labor demand function

$$Z_i(\omega_i; \omega_j) \equiv E_i^d(\omega_i; \omega_j) - E_i^s(\omega_i).$$

By Lemma 3 and Assumption 4, $E_i^d(\cdot; \omega_j)$ is continuous and strictly decreasing, with $\lim_{\omega_i \rightarrow 0^+} E_i^d = +\infty$ and $\lim_{\omega_i \rightarrow \infty} E_i^d = 0$. The supply schedule E_i^s is continuous and strictly increasing, satisfies $\lim_{\omega_i \rightarrow 0^+} E_i^s = 0$, and is bounded above by $N_i L_i$ because $H_i \leq 1$. Hence $Z_i(\cdot; \omega_j)$ is continuous and strictly decreasing, with a positive limit as $\omega_i \rightarrow 0^+$ and a negative value once E_i^d falls below E_i^s ; the intermediate value theorem therefore yields a unique conditional wage

$$\omega_i = \phi_i(\omega_j), \quad Z_i(\phi_i(\omega_j); \omega_j) = 0.$$

The next lemma guarantees that this root delivers an interior employment rate, so that the upper bound $H_i \leq 1$ never binds.

Lemma 4 (Interior employment). *Fix a configuration $d \in \mathcal{D}$ and a foreign wage ω_j , $j \neq i$. Let $\omega_i^d(\omega_j)$ denote the unique solution of*

$$E_i^d(\omega_i; \omega_j) = N_i L_i,$$

which exists and is unique because, by Assumption 4, $E_i^d(\cdot; \omega_j)$ is continuous and strictly decreasing with $\lim_{\omega_i \rightarrow 0^+} E_i^d = +\infty$ and $\lim_{\omega_i \rightarrow \infty} E_i^d = 0$; that is, $\omega_i^d(\omega_j)$ is the wage at which labor demand equals the entire labor force $N_i L_i$. If $a_i > \omega_i^d(\omega_j)$, then the conditional wage satisfies $\phi_i(\omega_j) \in [\omega_i^d(\omega_j), a_i)$, and hence the equilibrium employment rate

$$H_i^* = (\phi_i(\omega_j)/a_i)^{1/\beta_i} < 1,$$

so the upper bound $H_i \leq 1$ does not bind.

Proof. Throughout, ω_j is held fixed and the dependence of E_i^d and ω_i^d on it is suppressed.

Lower bound. At $\omega_i = \omega_i^d$ we have $E_i^d(\omega_i^d) = N_i L_i$, while labor supply is $E_i^s(\omega_i^d) = H_i(\omega_i^d) N_i L_i \leq N_i L_i$ because $H_i \leq 1$ by (8). Hence

$$Z_i(\omega_i^d) = N_i L_i - H_i(\omega_i^d) N_i L_i \geq 0.$$

Upper bound. At $\omega_i = a_i$ the wage curve (8) gives $H_i(a_i) = 1$, so labor supply equals $N_i L_i$. Since $a_i > \omega_i^d$ and E_i^d is strictly decreasing, $E_i^d(a_i) < E_i^d(\omega_i^d) = N_i L_i$, and therefore

$$Z_i(a_i) = E_i^d(a_i) - N_i L_i < 0.$$

Conclusion. Z_i is continuous and strictly decreasing, so its unique root $\phi_i(\omega_j)$ lies in $[\omega_i^d, a_i]$: it is weakly above ω_i^d because $Z_i(\omega_i^d) \geq 0$, and strictly below a_i because $Z_i(a_i) < 0$. Since $H_i(\cdot)$ is strictly increasing and $H_i(a_i) = 1$, the strict inequality $\phi_i(\omega_j) < a_i$ implies $H_i^* = H_i(\phi_i(\omega_j)) < 1$. $\square$

(ii) Two-country system. Equilibrium relative wages solve the pair of fixed-point conditions

$$\omega_h = \phi_h(\omega_f), \quad \omega_f = \phi_f(\omega_h),$$

where ϕ_i is the conditional wage function constructed in step (i). The absolute price level is pinned down by the numeraire normalization (1), so there is no residual scale indeterminacy, and the balance-of-payments condition (33) is implied by Walras' law once both labor markets clear (see the counting argument above). Existence follows from the continuity of ϕ_h and ϕ_f together with the boundary behavior established in step (i): by Lemma 4, each ϕ_i maps a compact interval of wages into itself, so Brouwer's fixed-point theorem applies. Local uniqueness follows because the Jacobian

$$J \equiv \frac{\partial(Z_h, Z_f)}{\partial(\ln \omega_h, \ln \omega_f)}$$

has a strictly dominant diagonal (Lemma 9 in Appendix A.1), which implies $\det J \neq 0$; the implicit function theorem then applies at every parameter vector outside the critical set Θ^c characterized in Section 2.2.3.

2.2.3 Existence and Generic Local Uniqueness

A competitive equilibrium in the sense of Definition 1 always exists and is generically locally unique. The argument proceeds in three steps.

Step 1: the configuration space is finite. The set $\mathcal{D}$ of organizational configurations (34) is finite, because chain length is bounded above by $\bar{n}_i \leq \bar{N}(\chi)$ and, for each admissible length, the feasible offshoring set $\mathcal{K}_i$ ranges over a finite collection of subsets of stages.

Step 2: conditional wages exist and are unique. Fix a configuration $d \in \mathcal{D}$. By Assumption 4 and Lemma 3, the excess labor demand function $Z_i(\cdot; d)$ is continuous and strictly decreasing, is positive as $\omega_i \rightarrow 0^+$ and becomes negative for ω_i large. Hence, by step (i) of the preceding characterization and Lemma 4, the conditional wage vector $\omega^*(d)$ exists, is unique, and delivers interior employment in both countries.

Step 3: the discrete choice admits a fixed point. Given $\omega^*(d)$, let each country solve its own configuration problem (13)–(17) and define the best-response correspondence

$$\mathcal{B}(d) \equiv \prod_{i \in \{h, f\}} \arg \min_{d'_i} F_i(d'_i; \omega^*(d)), \quad (37)$$

where F_i is country i 's terminal unit cost under configuration d'_i and the product is taken over countries, so that $\mathcal{B}$ maps the finite set $\mathcal{D}$ into itself. A configuration d^* is an equilibrium configuration if and only if $d^* \in \mathcal{B}(d^*)$. Lemma 11 in Appendix A.1 shows that $\mathcal{B}$ admits such a fixed point, which establishes existence; under the tie-breaking convention of Assumption 3(iii), attention is restricted to the smallest fixed point in the product order, so that d^* is uniquely selected.

Global uniqueness is neither claimed nor needed. Let Θ^c collect the parameter vectors $\theta \in \Theta$ at which two distinct configurations attain the same terminal cost for some country, that is, at which

$$\Delta_{i,n}(\theta) = 0 \quad \text{or} \quad \tau_i P_{j,n-1}(\theta) - P_{i,n-1}(\theta) = 0$$

for some $i \in \{h, f\}$ and some stage n , where $\Delta_{i,n}$ is the marginal gain from extending the chain to stage n and the second equation states indifference between domestic and offshore sourcing of stage n . Across Θ^c the discrete objects $\bar{n}_i$ and $\mathcal{K}_i$ jump, so aggregate excess demand is only piecewise continuous and multiple equilibria may coexist. Away from Θ^c ,

by contrast, d^* is locally constant, so the comparative statics with respect to N_i developed in Section 2.4 are well defined. This dichotomy is the formal counterpart of the empirical results: smooth intensive-margin responses of GVC participation to participation rates within cells, and discrete reorganizations of the chain – in chain length and in the set of imported stages – when the economy crosses a critical boundary.

Each function defining Θ^c is C^1 with a nonvanishing gradient in θ . Indeed, $\partial\Delta_{i,n}/\partial\omega_i \neq 0$ by Assumption 3(ii), and $\partial(\tau_i P_{j,n-1} - P_{i,n-1})/\partial\tau_i = P_{j,n-1} > 0$ because prices are strictly positive by Lemma 3. By the implicit-function theorem, each equation therefore defines a C^1 hypersurface of codimension-one in Θ . Since finitely many such equations are involved, Θ^c is a finite union of codimension-one hypersurfaces and hence a closed set of Lebesgue measure zero, so its complement is open and dense in Θ . On that complement the Jacobian J of step (ii) is nonsingular, and the implicit-function theorem delivers local uniqueness of the equilibrium; this is the sense in which the equilibrium is generically locally unique.

2.3 Welfare

This subsection derives the welfare measure implied by Definition 1, verifies that it is consistent with the income accounting of Section 2.2, and characterizes the gains from trade. The main result is that the model departs from the sufficient-statistic class of Arkolakis et al. (2012): the domestic expenditure share and the elasticity of substitution alone do not determine welfare, because trade moves both the wage ω_i and the employment rate H_i , and these two margins need not move in the same direction.

Before stating the measure, two conventions are recorded. First, all nominal magnitudes, including wages ω_i and producer prices p_i , are expressed in units of the numeraire fixed in (1), namely the internationally traded coordination service with $p^M \equiv 1$. Second, welfare is defined as a ratio of two numeraire-denominated magnitudes, so the normalization affects the level of neither numerator nor denominator in a way that matters: it cancels in every comparison reported below.

2.3.1 Real Income

Households are risk neutral over consumption and pool employment risk within the representative family, so that all members consume the same bundle irrespective of individual employment status. Per-capita real income is then a valid ordinal representation of household welfare, and no separate distributional correction is required.

Maximizing (2) subject to (3) yields the indirect utility $U_i = \omega_i E_i / P_{U_i}$, where E_i is aggregate employment, ω_i is the wage expressed in the numeraire fixed above, and the consumer price index P_{U_i} is the dual of (2), derived in (6). Using $E_i = H_i N_i L_i$ from (7), where H_i is the endogenous employment rate among participants and N_i the exogenous participation rate, and dividing by the fixed population L_i , welfare per capita is

$$W_i \equiv \frac{U_i}{L_i} = \frac{\omega_i E_i}{L_i P_{U_i}} = \frac{\omega_i H_i N_i}{P_{U_i}}, \quad i \in \{h, f\}. \quad (38)$$

Normalizing by L_i is not innocuous for the comparative statics below. Because the employment margin H_i is the object of interest, aggregate real income $\omega_i E_i / P_{U_i}$ would confound it with a pure population scale effect; per-capita real income isolates the margin. Since N_i and L_i are both fixed, W_i varies across equilibria and across regimes only through ω_i , H_i and P_{U_i} .

Equation (38) makes the three channels through which trade affects welfare explicit: (i) a price effect, operating through the delivered import price $\tau_i p_j$ in P_{U_i} ; (ii) a wage effect, operating through ω_i ; and (iii) an employment effect, operating through H_i . The coordination wedge χ enters welfare exclusively through producer prices: by the stage-price recursion (14) of Section 2.1.3, χ raises $P_{i, \bar{n}_i}$, hence P_{X_i} in (23) and p_i in (26), so that its welfare incidence is fully embedded in P_{U_i} . Throughout, $\sigma > 1$ denotes the elasticity of substitution across final varieties in (2); it is distinct from the intermediate-input elasticity $\sigma_{X_i} = 1 + \alpha_i$ introduced in Section 2.1.5.

Income accounting. Equation (38) measures real income because its numerator is household income per capita. Assumption 1 rules out household ownership of the coordination endowment, and firms earn zero profits, so $I_i = \omega_i E_i$ from (4). Nonsatiation makes the budget constraint (3) bind, so in any equilibrium of Definition 1,

$$p_i Q_{ii} + \tau_i p_j Q_{ij} = \omega_i E_i, \quad j \neq i, \quad (39)$$

where Q_{ii} and Q_{ij} denote country i 's consumption of the domestic and the imported final variety, respectively.

Coordination spending does not appear on either side of (39). Summing the zero-profit conditions across stages and the final-good sector gives

$$\omega_i E_i = \lambda_i^{\text{fin}} p_i F_i + \sum_{n=1}^{\bar{n}_i} \lambda_{i,n} V_{i,n} X_{i,n},$$

where λ_i^{fin} and $\lambda_{i,n}$ are the labor shares of, respectively, final-good and stage- n unit cost, and $V_{i,n}$ and $X_{i,n}$ are the stage- n unit cost and output defined in Section 2.1.3. The coordination component $\chi(n-1)X_{i,n}$ of stage revenue accrues to the world coordination sector and therefore generates no domestic labor demand. Imports of coordination services D_i^M are financed by net goods exports through (33) rather than out of household income; their real resource cost is passed into producer prices by the stage-price recursion (14), so that coordination affects welfare only through p_i in P_{Ui} . Relaxing (33) to allow an exogenous imbalance $EX_i - IM_i \neq 0$ would replace the numerator of (38) by $\omega_i E_i + (EX_i - IM_i)$, leaving the three welfare channels identified above unchanged.

2.3.2 Gains from Trade and the Autarky Benchmark

Goods autarky. The benchmark shuts down trade in goods but not in coordination services. It is the equilibrium of Definition 1 with $\tau_h = \tau_f = \infty$, so that $\mathcal{K}_h = \mathcal{K}_f = \emptyset$ in (13) and neither the final good nor any stage-level intermediate is traded, while coordination services continue to trade at $p^M = 1$. Keeping the coordination market open

is necessary for comparability: closing it would leave χ and the numeraire (1) undefined, so that the two regimes could not be placed on a common scale. Since $\sigma > 1$, the import term $(\tau_i p_j)^{1-\sigma}$ vanishes as $\tau_i \rightarrow \infty$ and the domestic expenditure share s_{ii} converges to one, so that the consumer price index collapses to $P_{U_i}^{\text{aut}} = p_i^{\text{aut}}$. It is the trade-cost limit, not the value of σ , that eliminates imports; the restriction $\sigma > 1$ only ensures that the import share declines monotonically in τ_i .

Without goods trade, the balance of payments (33) has a zero left-hand side while $D_i^M > 0$, so coordination imports cannot be financed by exports. We therefore finance them with a lump-sum transfer from the rest of the world equal to D_i^M , which is the limit of (33) when the export revenue that pays for coordination is replaced by a transfer. The transfer is received by firms and spent entirely on coordination inputs, and households own no coordination endowment, so (4) and (38) retain the form derived above: coordination affects autarky welfare only through p_i^{aut} , exactly as it affects P_{U_i} under trade. Autarky welfare per capita is therefore

$$W_i^{\text{aut}} = \frac{\omega_i^{\text{aut}} H_i^{\text{aut}} N_i}{p_i^{\text{aut}}}, \quad i \in \{h, f\}, \quad (40)$$

where N_i carries no regime superscript because participation is exogenous, as in (38).

Two features make the comparison well posed. First, the chain length is endogenous in the benchmark: $\bar{n}_i^{\text{aut}}$ solves (17) at the autarky wage along a fully domestic chain and generally differs from the observed $\bar{n}_i$, so that only the primitives $(a_i, \beta_i, N_i, L_i, c, \chi, \sigma, \alpha_i)$ are held fixed across regimes. Second, $(\omega_i^{\text{aut}}, H_i^{\text{aut}})$ solve (8) and (32) at the autarky organizational configuration d^{aut} , so that the labor market adjusts across regimes rather than being held at its trade-equilibrium allocation.

Gains from trade. Let $\text{Gain}_i \equiv W_i/W_i^{\text{aut}} - 1$. Substituting (38) and (40), noting that the exogenous participation rate N_i cancels, and using the CES identity $P_{U_i} = p_i s_{ii}^{1/(\sigma-1)}$

implied by (6) together with $P_i^{\text{aut}} = p_i^{\text{aut}}$,

$$\frac{W_i}{W_i^{\text{aut}}} = \underbrace{s_{ii}^{-\frac{1}{\sigma-1}}}_{\text{ACR term}} \times \underbrace{\frac{\omega_i/\omega_i^{\text{aut}}}{p_i/p_i^{\text{aut}}}}_{\text{real producer wage}} \times \underbrace{\frac{H_i}{H_i^{\text{aut}}}}_{\text{employment margin}} = s_{ii}^{-\frac{1}{\sigma-1}} \Lambda_i, \quad i \in \{h, f\}, \quad (41)$$

where the last two terms are collected in

$$\Lambda_i \equiv \frac{\omega_i/\omega_i^{\text{aut}}}{p_i/p_i^{\text{aut}}} \cdot \frac{H_i}{H_i^{\text{aut}}}, \quad \ln \Lambda_i = (\ln \omega_i - \ln \omega_i^{\text{aut}}) - (\ln p_i - \ln p_i^{\text{aut}}) + (\ln H_i - \ln H_i^{\text{aut}}). \quad (42)$$

The first term in (41) is the sufficient statistic of Arkolakis et al. (2012), and Λ_i is the change in real factor income per capita, deflated by the domestic producer price rather than by the consumer price index. In the ACR class $\Lambda_i = 1$: the wage is the numeraire, the producer price moves with it, and the labor endowment is supplied inelastically, so that s_{ii} and σ exhaust the gains. Here both remaining margins move. Trade can raise the producer wage by shortening the domestic segment of the chain, and it shifts the employment rate through (32); the sign of each effect is established in Section 2.4. The domestic expenditure share is therefore not a sufficient statistic for welfare in this model, and $s_{ii}^{-1/(\sigma-1)}$ understates the gains whenever $\Lambda_i > 1$ and overstates them whenever $\Lambda_i < 1$. Section 2.3.3 states the condition under which the two coincide.

2.3.3 Departure from the Sufficient-Statistic Class

Welfare gains are not determined by (s_{ii}, σ) . By (41), $W_i/W_i^{\text{aut}} = s_{ii}^{-1/(\sigma-1)}$ as in Arkolakis et al. (2012) if and only if $\Lambda_i = 1$, that is, if and only if trade leaves real factor income per capita, deflated by the domestic producer price, unchanged:

$$\frac{\omega_i H_i}{p_i} = \frac{\omega_i^{\text{aut}} H_i^{\text{aut}}}{p_i^{\text{aut}}}, \quad i \in \{h, f\}, \quad (43)$$

where the exogenous participation rate N_i has again cancelled from both sides.

Two conditions are jointly sufficient for (43): a frictionless labor market, $\beta_i = 0$, which by (8) and (32) delivers $H_i = H_i^{\text{aut}} = 1$ and hence a wage pinned down by labor demand alone; and invariance of the domestic producer price relative to the wage, $\omega_i/p_i =$

$\omega_i^{\text{aut}}/p_i^{\text{aut}}$, which holds when the chain length is fixed at $\bar{n}_i = \bar{n}_i^{\text{aut}}$ and no stage is offshored. Both are needed: the second alone leaves the employment ratio in Λ_i , and the first alone leaves the real producer wage free to move. For $\beta_i > 0$ the first fails, and (43) then holds only on a measure-zero set of parameters, so that (s_{ii}, σ) are not sufficient statistics for welfare in this model.

The economic content is that the sufficient-statistic result requires trade not to affect real factor income measured in producer prices. With frictional labor markets that requirement fails by construction: the wage curve (8) ties ω_i to H_i , and (32) ties H_i to sectoral labor demand, which trade shifts. In the present model, trade lowers intermediate-input prices by widening the set of accessible sourcing margins in (13); final-goods output expands and labor demand in (32) rises, which moves both H_i and, through (8), ω_i . The direction and relative magnitude of the two movements, and hence the sign of $\ln \Lambda_i$, are established in Section 2.4. Both movements enter Λ_i ; neither is captured by s_{ii} .

2.3.4 Frictionless Trade and the Amplification Condition

Consider free trade, $\tau_h = \tau_f = 1$, between symmetric countries, $a_h = a_f$, $\beta_h = \beta_f$, $N_h = N_f$, $L_h = L_f$, and $\alpha_h = \alpha_f$. Symmetry implies $P_{h,n} = P_{f,n}$ at every stage, so the tie-breaking convention of Section 2.2.2 gives $\mathcal{K}_i = \emptyset$, and the chain length solving (17) coincides with its autarky counterpart evaluated at the same wage. Writing $p_h = p_f \equiv p^{\text{FT}}$, and noting that each country consumes both final varieties, the price index (6) and the domestic expenditure share become

$$P_{Uh}^{\text{FT}} = p^{\text{FT}} 2^{\frac{1}{1-\sigma}}, \quad s_{hh}^{\text{FT}} = \frac{1}{2}. \quad (44)$$

Trade therefore operates through two channels only. First, the set of available final varieties doubles, which lowers the price index by the factor $2^{1/(1-\sigma)} < 1$ at a given producer price. Second, factor income adjusts through Λ_h , because the reallocation of production across borders shifts labor demand in (32). Production costs at a given wage are unchanged in

the symmetric free-trade equilibrium itself, since no stage is sourced abroad there; the cost channel is nevertheless active along the path to free trade, where $\mathcal{K}_i \neq \emptyset$ for intermediate values of τ_i .

To characterize the second channel, define the trade-cost elasticities of the employment rate and of the domestic producer price,

$$\eta_H \equiv -\frac{d \ln H_h}{d \ln \tau_h}, \quad \eta_p \equiv -\frac{d \ln p_h}{d \ln \tau_h}, \quad i \in \{h, f\}, \quad (45)$$

both evaluated along the symmetric liberalization path, with $\eta_H > 0$ because a lower τ_h raises labor demand in (32), while the sign of η_p is left unrestricted. Differentiating the wage curve (8) gives $d \ln \omega_h = \beta_h d \ln H_h$, so (42) implies

$$-\frac{d \ln \Lambda_h}{d \ln \tau_h} = (1 + \beta_h) \eta_H - \eta_p. \quad (46)$$

Liberalization raises real factor income per capita, deflated by the domestic producer price, if and only if

$$(1 + \beta_h) \eta_H > \eta_p, \quad (47)$$

that is, if and only if the employment response, amplified by the wage-curve elasticity β_h , dominates the pass-through of trade costs into the domestic producer price.

If (47) holds along the whole liberalization path, then $\Lambda_h > 1$ and the gains exceed $(s_{hh}^{\text{FT}})^{-1/(\sigma-1)} = 2^{1/(\sigma-1)}$; if the inequality is reversed throughout, the income channel offsets the price channel and the gains fall short of that benchmark. Two limits delimit the condition. With a frictionless labor market, $\beta_h = 0$ and an inelastic employment rate, $\eta_H = 0$, the right-hand side of (46) reduces to $-\eta_p$, so the benchmark of Section 2.3.3 obtains exactly when in addition $\eta_p = 0$, which is the second condition in (43). At the other extreme, when η_p is large, liberalization reallocates production abroad and raises the cost of the stages retained at home, so that the labor-market channel attenuates rather than amplifies the gains. Whether (47) holds is thus a quantitative question about β_h and η_H , which is why both are estimated rather than calibrated in Section 3.

2.4 Main Theoretical Results

This subsection derives four propositions concerning the core indicators of global value chain (GVC) organization: forward and backward participation, GVC position, chain length, and the split between domestic and offshored production stages. Each proposition is accompanied by a corollary describing how trade costs moderate the effect, which delivers the interaction terms estimated in Section 3.

The central mechanism operates in two steps. First, a higher participation rate N_h lowers the equilibrium wage ω_h , through the interaction of the wage curve (8) with the labor-market clearing condition (32). Second, a lower wage reduces the cost of domestically produced stages relative both to imported stages and to the wage-invariant coordination overhead χ , which reorganizes sourcing along two margins and shortens the chain. Since N_h is exogenous and L_h is fixed, these comparative statics are with respect to a primitive of the model rather than to an equilibrium object.

Throughout, $\varepsilon_{h,n} \in (0, 1]$ denotes the elasticity of the stage- n price $P_{h,n}$ with respect to the wage ω_h , as established in Lemma 3; all other elasticities are denoted by η with a subscript identifying the variable in the numerator. Unless stated otherwise, all comparative statics are evaluated at parameter vectors in $\Theta \setminus \Theta^c$, where by Section 2.2.3 the equilibrium is locally unique and the organizational configuration d^* is locally constant, and the foreign country is held at fixed primitives $(a_f, \beta_f, N_f, L_f, \alpha_f)$.

We further distinguish the intensive margins of Propositions 1–2, evaluated where d^* is locally constant, from the *discrete* reorganization of Proposition 3, which by definition involves a change in d^* and is therefore stated as a monotone comparative static across configurations rather than as a partial derivative.

Assumption 5 (Small open economy for comparative statics). When evaluating the comparative statics of Section 2.4, the foreign wage ω_f and the foreign configuration $(\bar{n}_f, \mathcal{K}_f)$ are held invariant to N_h and ω_h . Foreign stage prices $P_{f,n}$ are held invariant for all

stages that use no home-produced input, either directly or indirectly, that is, for all $n < \min \mathcal{K}_f$; for $n \geq \min \mathcal{K}_f$ the price $P_{f,n}$ responds to ω_h only through the imported inputs $\hat{P}_{f,m} = \tau_f P_{h,m}$ with $m \in \mathcal{K}_f$, according to the recursion (14). If $\mathcal{K}_f = \emptyset$, all foreign stage prices are held invariant.

Assumption 5 restricts only the comparative-statics exercise, not the equilibrium concept of Section 2.2.2. It shuts down the general-equilibrium feedback from home to foreign wages, which Lemma 10 shows to be of second order relative to the direct effect,

$$\left| \frac{d \ln \omega_f}{d \ln N_h} \right| < \left| \frac{d \ln \omega_h}{d \ln N_h} \right| \equiv |\kappa_\omega|,$$

while retaining the direct cost pass-through from home stage prices to those foreign stages that source from home. The latter is required for forward participation to be well defined.

2.4.1 The Wage Channel

Lemma 5 (Wage and price effects of participation). *Fix a configuration $d \in \mathcal{D}$ with $\beta_h > 0$, and let Assumption 5 hold. Let $\eta_h \equiv -\partial \ln E_h^d / \partial \ln \omega_h > 0$ denote the aggregate labor-demand elasticity of Assumption 4 (see (A.5) in Appendix A.1). Then*

$$\kappa_\omega \equiv \frac{d \ln \omega_h}{d \ln N_h} = -\frac{\beta_h}{1 + \beta_h \eta_h} < 0, \quad \frac{d \ln H_h}{d \ln N_h} = \frac{\kappa_\omega}{\beta_h} = -\frac{1}{1 + \beta_h \eta_h} \in (-1, 0), \quad (48)$$

and

$$\frac{d \ln P_{h,n}}{d \ln N_h} = \varepsilon_{h,n} \kappa_\omega < 0 \quad \text{for every } n \leq \bar{n}_h, \quad (49)$$

while the coordination overhead $\chi(n-1)$ and the foreign price path $\{P_{f,n}\}_{n \leq \bar{n}_f}$ are unaffected.

Proof. Log-differentiate the wage curve (8) to obtain $d \ln \omega_h = \beta_h d \ln H_h$, and the labor-market clearing condition (32), written as $E_h^d(\omega_h) = H_h N_h L_h$, to obtain $-\eta_h d \ln \omega_h = d \ln H_h + d \ln N_h$, where L_h is fixed and η_h is defined in (A.5). Substituting $d \ln H_h = d \ln \omega_h / \beta_h$ and solving gives $d \ln \omega_h (\eta_h + 1/\beta_h) = -d \ln N_h$, which is (48); the signs follow from $\beta_h > 0$

and $\eta_h > 0$, the latter guaranteed by Assumption 4 and established from primitives in Lemma 7, and the bound $d \ln H_h / d \ln N_h > -1$ follows because $\beta_h \eta_h > 0$. Equation (49) then follows from Lemma 3, which gives $\varepsilon_{h,n} \in (0, 1]$. The overhead is denominated in the numeraire (1) and is therefore invariant to ω_h , and the foreign block is unchanged under Assumption 5 because N_h enters no foreign primitive. $\square$

Lemma 5 isolates the two forces behind all results below. A rise in N_h lowers the entire domestic stage-price path relative to the imported alternative $\tau_h P_{f,n}$, which is invariant to ω_h by Assumption 5 for the stages that do not embody home inputs; and it lowers the wage relative to the wage-invariant coordination cost, that is, it raises χ/ω_h . Because $d \ln H_h / d \ln N_h \in (-1, 0)$, aggregate employment $H_h N_h L_h$ still rises with N_h , so the participation shock expands employment while compressing the wage.

Condition 1 (Trade-cost complementarity in chain length). *For every $n \leq \bar{n}_h$, the threshold wage $\omega_h^{(n)}(\tau_h)$ defined by $\Delta_h(n; \omega_h, \tau_h) = 0$ is weakly decreasing in τ_h . A sufficient condition is*

$$\frac{\partial^2 \Delta_h(n; \omega_h, \tau_h)}{\partial \ln \omega_h \partial \ln \tau_h} \geq 0 \quad \text{and} \quad \frac{\partial \Delta_h(n; \omega_h, \tau_h)}{\partial \ln \tau_h} \geq 0 \quad \text{for all } n \leq \bar{n}_h.$$

Condition 2 (Trade-cost complementarity in sourcing). *For every $n \leq \bar{n}_h$,*

$$\frac{\partial^2 \ln(P_{h,n}/\tau_h P_{f,n})}{\partial \ln \omega_h \partial \ln \tau_h} \leq 0.$$

2.4.2 Proposition 1: Forward and Backward GVC Participation

Following Koopman et al. (2014) and Wang et al. (2013), forward and backward GVC participation are monotone transformations of the share of domestic intermediate output absorbed abroad and of the share of intermediate input expenditure sourced abroad:

$$\text{GVCpar}_{f,h} = \Psi(A_h), \quad A_h \equiv \frac{P_{h,\bar{n}_h} X_{fh} + \sum_{n \in \mathcal{K}_f} P_{h,n} M_{f,n}}{\sum_{n=1}^{\bar{n}_h} P_{h,n} X_{h,n}}, \quad \Psi' > 0, \quad (50)$$

$$\text{GVCpar}_{b,h} = \Phi(B_h), \quad B_h \equiv \frac{\tau_h P_{f,\bar{n}_f} X_{hf} + \sum_{n \in \mathcal{K}_h} \tau_h P_{f,n} M_{h,n}}{P_{Xh} X_h + \sum_{n=2}^{\bar{n}_h} (1 - \lambda_{h,n}) V_{h,n} X_{h,n}}, \quad \Phi' > 0, \quad (51)$$

where X_{fh} denotes home final-good sales to the foreign country, X_{hf} foreign final-good sales to home, and $M_{i,n}$ the quantity of the stage- n input imported by country i , so that the numerator of A_h collects all home output absorbed abroad and the numerator of B_h all home expenditure on foreign-produced goods, valued at delivered prices $\tau_h P_{f,n}$.

Both A_h and B_h are ratios of expenditures, so numerator and denominator are measured in the same units and, being homogeneous of degree zero in the scale of production, the comparative statics do not depend on that scale. Because Ψ and Φ are strictly increasing and differentiable, the signs derived below for A_h and B_h carry over to the observed indices.

Proposition 1 (Participation). *An increase in N_h lowers backward participation and raises forward participation,*

$$\frac{d \text{GVCpar}_{b,h}}{dN_h} < 0, \quad \frac{d \text{GVCpar}_{f,h}}{dN_h} > 0. \quad (52)$$

Proof. Both A_h and B_h are ratios of expenditures measured in the same units, so a common proportional change in all stage outputs $\{X_{h,n}\}_n$ leaves them unchanged: the scale terms in numerator and denominator cancel, and the signs in (52) are governed by relative prices alone. It therefore suffices to sign the response of the underlying expenditure shares. Lemma 3 implies that a rise in N_h lowers ω_h and hence the entire domestic price path $\{P_{h,n}\}_n$, while under Assumption 5 the delivered import prices $\tau_h P_{f,n}$ and the foreign wage ω_f are unchanged; consequently $P_{h,n}/(\tau_h P_{f,n})$ falls at every stage.

Consider B_h . Two margins operate. On the *intensive* margin, the CES demands (24) imply that the imported expenditure share of the stage- n composite satisfies

$$\frac{d \ln(\tau_h P_{f,n-1} M_{h,n} / \hat{P}_{h,n-1} X_{h,n})}{d \ln P_{h,n-1}} = (\sigma_{Xh} - 1) s_{h,n} > 0, \quad (53)$$

where $s_{h,n}$ is the domestic expenditure share defined in (35) and $\sigma_{Xh} = 1 + \alpha_h > 1$; hence the fall in domestic stage prices lowers every imported share, and the same relation applied to the terminal stage lowers $\tau_h P_{f,\bar{n}_f} X_{hf} / P_{Xh} X_h$. On the *extensive* margin, since

$P_{h,n}/(\tau_h P_{f,n})$ falls at every stage, the sourcing criterion $\tau_h P_{f,n} < P_{h,n}$ in (13) is satisfied for a weakly smaller set of stages: $\mathcal{K}_h$ contracts weakly, and every retained imported stage carries a weakly lower expenditure weight. Both margins lower imported expenditure relative to the common scale, so B_h falls and $\Phi' > 0$ delivers the first inequality.

For A_h , the same price movement makes domestic stages weakly cheaper for foreign buyers at unchanged τ_f and $\{P_{f,n}\}_n$, so $\mathcal{K}_f$ weakly expands by (13). On the intensive margin, the foreign counterparts of (24) and (53), with $\sigma_{Xf} = 1 + \alpha_f > 1$, raise the share of foreign stage expenditure allocated to X_{fh} and to $\{M_{f,n}\}_{n \in \mathcal{K}_f}$, while the CES final-goods demand (5) with $\sigma > 1$ raises foreign absorption of the terminal domestic good. Since the scale of domestic stage output enters numerator and denominator of A_h proportionally, these share movements are not offset, so A_h rises and $\Psi' > 0$ delivers the second inequality. Lemma 17 in Appendix A.5 reports the corresponding elasticities,

$$d \ln B_h / d \ln N_h = -(\sigma_{Xh} - 1)(1 - \bar{s}_h) \bar{\varepsilon}_h \kappa_\omega < 0$$

and

$$d \ln A_h / d \ln N_h = -(\sigma - 1)(1 - A_h) \varepsilon_h^t \kappa_\omega > 0.$$

□

Corollary 0.1 (Trade-cost moderation of participation). *Suppose Condition 2 holds. Then the effects in (52) are weakly stronger at higher trade costs:*

$$\frac{\partial^2 \text{GVCpar}_{b,h}}{\partial N_h \partial \ln \tau_h} \leq 0, \quad \frac{\partial^2 \text{GVCpar}_{f,h}}{\partial N_h \partial \ln \tau_h} \geq 0, \quad (54)$$

with strict inequalities whenever the intensive margin is operative, that is whenever $s_{h,n} \in (0, 1)$ for some $n \leq \bar{n}_h$.

Sketch. Two channels operate, and they differ in what they require. The *intensive*-margin channel is unconditional: a higher τ_h raises $\tau_h P_{f,n-1}/\hat{P}_{h,n-1}$ and hence the domestic expenditure share $s_{h,n}$ at every stage, so the coefficient $(\sigma_{Xh} - 1)s_{h,n}$ in (53) rises and a given

proportional wage decline produces a larger fall in the imported share; by (A.15) the same holds for the export-share elasticity governing A_h . The extensive-margin channel is conditional: a higher τ_h raises $\tau_h P_{f,n}/P_{h,n}$ at every stage, but whether this makes the sourcing margin more wage sensitive depends on the cross-partial $\partial^2 \ln(P_{h,n}/\tau_h P_{f,n})/\partial \ln \omega_h \partial \ln \tau_h$, which Condition 2 restricts to be weakly negative. Under that condition a given wage decline induces a weakly larger contraction of $\mathcal{K}_h$ and a weakly larger expansion of $\mathcal{K}_f$, so both channels reinforce the signs of Proposition 1. Appendix A.9 gives the formal argument. $\square$

2.4.3 Proposition 2: GVC Position

Define the GVC position index as the log difference between the forward and backward measures,

$$\text{POS}_h \equiv \ln(1 + A_h) - \ln(1 + B_h), \quad (55)$$

with A_h and B_h from (50)–(51). We use the explicit functional form (55) rather than an arbitrary increasing transformation, because the difference of two arbitrary monotone transformations does not in general preserve the sign of the underlying comparative statics.

Proposition 2 (GVC Position). *An increase in N_h moves country h to a more upstream GVC position,*

$$\frac{d\text{POS}_h}{d \ln N_h} = \kappa_\omega \left[\frac{A_h}{1 + A_h} \kappa_A - \frac{B_h}{1 + B_h} \kappa_B \right] > 0, \quad (56)$$

where $\kappa_A \equiv \partial \ln A_h / \partial \ln \omega_h < 0$ and $\kappa_B \equiv \partial \ln B_h / \partial \ln \omega_h > 0$.

Proof. Differentiating (55) and writing each derivative in elasticity form gives (56). The signs of κ_A and κ_B follow from the proof of Proposition 1, and $\kappa_\omega < 0$ by Lemma 3. Since $A_h, B_h > 0$, the two weights $A_h/(1 + A_h)$ and $B_h/(1 + B_h)$ lie in $(0, 1)$, so the bracket is strictly negative; multiplied by $\kappa_\omega < 0$ it yields a strictly positive expression. Outside Θ^c the organizational configuration d^* is locally constant by Lemma 10, so no discrete reorganization interferes with the differentiation. $\square$

Corollary 0.2 (Trade-Cost Moderation of GVC Position). *Under Condition 2, the positive upstream effect of N_h on POS_h is amplified by trade costs:*

$$\frac{\partial^2 \text{POS}_h}{\partial \ln N_h \partial \ln \tau_h} > 0. \quad (57)$$

The moderation result holds strictly whenever $\mathcal{K}_h \neq \emptyset$ or $s_{h,n} \in (0, 1)$ for some $n \leq \bar{n}_h$, with no weak inequality ambiguity present for the GVC position margin.

Sketch. Higher trade costs τ_h raise κ_B and lower κ_A under Condition 2, rendering the bracketed term in (56) more negative. Combined with $\kappa_\omega < 0$, this strengthens the positive marginal effect of N_h on upstream positioning. Unlike the GVC participation margins which allow locally inactive margins and weak inequalities, the opposing signed responses of forward and backward participation ensure the position effect is always strictly positive whenever either the extensive or intensive GVC margin is active. $\square$

The economic content of Proposition 2 is that a rise in N_h lowers the domestic wage, which makes home-produced stages cheaper relative both to imported stages and to foreign competitors, so home supplies more intermediates to foreign downstream producers and buys fewer intermediates from abroad. The country therefore specializes in earlier, upstream stages of the global value chain. The unambiguous sign in (56) arises because forward and backward participation respond to a common wage-driven price movement, requiring no additional restrictions on the relative magnitudes of κ_A and κ_B .

2.4.4 Proposition 3: Length of the Production Chain

Recall from (18) that the incremental value of an additional stage is $\Delta_h(n) = [V_{h,n+1} - V_{h,n}] + \chi$, and that $\bar{n}_h$ solves (17), equivalently (19) under Assumption 3. Propositions 1 and 2 are intensive-margin statements evaluated on $\Theta \setminus \Theta^c$, where d is locally constant. The chain-length response is, by construction, a *discrete* margin: it operates through crossings

of the critical set Θ^c of Section 2.2.3(iii). By (20) and Lemma 1,

$$\Delta_h(n) = -\omega_h(v_{h,n} - v_{h,n+1}) + \chi, \quad \frac{\partial \Delta_h(n)}{\partial \omega_h} = -(v_{h,n} - v_{h,n+1}) < 0, \quad (58)$$

so $\Delta_h(n)$ is strictly decreasing in ω_h for each n and crosses zero at most once. Let $\omega_h^{(n)}(\tau_h)$ denote that crossing point, defined by $\Delta_h(n; \omega_h, \tau_h) = 0$, and let $\Omega_h^* \equiv \{\omega_h^{(n)}(\tau_h)\}_{n \leq \bar{n}_h^{\max}}$, a finite set by Lemma 14.

Proposition 3 (Chain length: threshold reorganization). *The optimal chain length $\bar{n}_h$ is weakly decreasing in N_h ,*

$$N'_h > N_h \implies \bar{n}_h(N'_h, \tau_h) \leq \bar{n}_h(N_h, \tau_h), \quad (59)$$

with strict inequality if and only if the induced wage change $\kappa_\omega \ln(N'_h/N_h)$ carries ω_h across at least one threshold $\omega_h^{(n)}(\tau_h)$; absent such a crossing (59) holds with equality and no chain-length adjustment occurs. Consequently observed average production length, a strictly increasing transformation G of $\bar{n}_h$, satisfies $\partial \text{PLY}_h / \partial N_h = G'(\bar{n}_h) \partial \bar{n}_h / \partial N_h \leq 0$.

Proof. The second expression in (58) verifies the hypothesis of Assumption 3, so a rise in ω_h shifts the map $n \mapsto \Delta_h(n)$ pointwise downward and the smallest n at which it turns nonnegative weakly rises; by Topkis' theorem this is $\partial \bar{n}_h / \partial \omega_h \geq 0$, as recorded in Lemma 14. By Lemma 5, $\kappa_\omega < 0$, so $N'_h > N_h$ implies $\omega_h(N'_h) < \omega_h(N_h)$ and hence $\partial \bar{n}_h / \partial N_h \leq 0$, which is (59). Since $\Delta_h(n; \cdot)$ is continuous and strictly decreasing in ω_h for each n , and $\chi(n-1)$ caps the relevant range of n at a finite $\bar{n}_h^{\max}$ uniformly on compacta, Ω_h^* is finite; off Ω_h^* all inequalities in (19) are strict, so $\bar{n}_h$ is locally constant and declines only by discrete jumps at Ω_h^* . The final claim follows from $G' > 0$. $\square$

The interpretation follows directly from (58). Fragmentation is worthwhile because splitting production raises stage productivity and economizes on labor, an absolute saving worth $\omega_h(v_{h,n} - v_{h,n+1})$, while each additional stage costs χ in coordination services whose price is fixed by the numeraire (1). Equivalently, a lower wage raises the ratio χ/ω_h of

the wage-invariant overhead to the wage-proportional saving. A higher participation rate lowers ω_h and hence the value of economizing on labor, while leaving the coordination penalty unchanged; the chain is consolidated into fewer, less specialized stages. This is why the mechanism requires the additive, wage-invariant overhead of Assumption 1: under a Hicks-neutral coordination cost the ratio $\Delta_h(n)/\omega_h$ would be wage-invariant and $\bar{n}_h$ would not respond to labor supply at all.

Corollary 0.3 (Trade-cost moderation of chain length). *Suppose Condition 2 holds. Then*

$$\frac{\partial^2 \bar{n}_h}{\partial N_h \partial \ln \tau_h} \leq 0, \quad (60)$$

that is, the chain shortening induced by a rise in N_h is weakly stronger when trade costs are higher: for $N'_h > N_h$ and $\tau'_h > \tau_h$,

$$\bar{n}_h(N'_h, \tau'_h) \leq \bar{n}_h(N'_h, \tau_h) \leq \bar{n}_h(N_h, \tau_h). \quad (61)$$

Absent Condition 2 the first inequality in (61) holds with equality and no chain-length amplification arises; amplification along this margin is therefore a conditional prediction.

Sketch. The threshold wage $\omega_h^{(\bar{n}_h)}(\tau_h)$ solves $\Delta_h(\bar{n}_h; \omega_h, \tau_h) = 0$ and satisfies $\partial \Delta_h(n)/\partial \omega_h = \varsigma_{h,n} < 0$ by (58). A higher τ_h raises the effective input price $\hat{P}_{h,n-1}$ at every stage that would otherwise be imported, by (12), which raises the wage-elastic component of $V_{h,n}$ relative to the wage-invariant overhead and hence, under Condition 2, weakly steepens the wage sensitivity of the stage margin, $\partial^2 \Delta_h(n)/\partial \ln \omega_h \partial \ln \tau_h \geq 0$. Since $\kappa_\omega < 0$ by Lemma 5, a given rise in N_h lowers ω_h and shifts Δ_h upward pointwise; the steeper the sensitivity, the larger that shift, so the threshold in (19) is crossed for a weakly larger set of stages and $\omega_h^{(n)}(\tau_h)$ is weakly decreasing in τ_h . Equation (61) then follows from Topkis' theorem exactly as in (59). Because $\bar{n}_h$ is integer valued, both inequalities are weak and strict only at crossings of Ω_h^* . The formal argument is in Appendix A.9. $\square$

2.4.5 Proposition 4: Domestic versus Global Production Stages

Decompose the chain into internationally fragmented and domestically retained stages, with $\mathcal{K}_h \subseteq \{1, \dots, \bar{n}_h - 1\}$ from (13),

$$\bar{n}_h = \underbrace{|\mathcal{K}_h|}_{n_h^G \text{ (global stages)}} + \underbrace{(\bar{n}_h - |\mathcal{K}_h|)}_{n_h^D \text{ (domestic stages)}}. \quad (62)$$

Proposition 4 (Domestic versus global stages). *Let $N'_h > N_h$, evaluate both allocations outside the critical set Θ^c , and write $\Delta x \equiv x(N'_h) - x(N_h)$. An increase in N_h weakly reduces the number of internationally fragmented stages,*

$$\Delta n_h^G \leq 0, \quad (63)$$

whereas the effect on the number of domestically retained stages is the sum of two opposing forces and its sign is not determined by theory,

$$\underbrace{\Delta n_h^D}_{\text{domestic stages}} = \underbrace{\Delta \bar{n}_h}_{\leq 0 \text{ (chain shortening)}} - \underbrace{\Delta n_h^G}_{\leq 0 \text{ (reduced fragmentation)}}, \quad (64)$$

so that the two channels enter with opposite signs. The domestic component declines if and only if the chain-shortening effect dominates the reduction in fragmentation, that is, if and only if $|\Delta \bar{n}_h| > |\Delta n_h^G|$. The domestic share, by contrast, responds unambiguously,

$$\Delta \left(1 - \frac{|\mathcal{K}_h|}{\bar{n}_h} \right) \geq 0, \quad \Delta \left(\frac{|\mathcal{K}_h|}{\bar{n}_h} \right) \leq 0. \quad (65)$$

All increments are integer or step valued and vanish except at the finitely many crossings collected in Ω_h^ and in the sourcing indifference loci of Θ^c .*

Proof. Inequality (63) combines two channels. First, hold $\bar{n}_h$ fixed. By Lemma 16, a rise in N_h lowers every domestic stage price, $d \ln P_{h,n} / d \ln N_h = \varepsilon_{h,n} \kappa_\omega < 0$, while under Assumption 5 the imported alternatives $\tau_h P_{f,n}$ are unchanged; hence $P_{h,n} / (\tau_h P_{f,n})$ falls at every stage and the sourcing criterion (13) is satisfied for a weakly smaller set of stages,

so $\mathcal{K}_h(N'_h) \subseteq \mathcal{K}_h(N_h)$, as in Step 1 of the proof of Proposition 1. Second, by Proposition 3 the chain weakly shortens, and since $\mathcal{K}_h \subseteq \{1, \dots, \bar{n}_h - 1\}$, any stage index at or above the new terminal stage also leaves $\mathcal{K}_h$. Both channels weakly lower $|\mathcal{K}_h|$, which gives (63). Equation (64) is the identity (62) in first differences, with $\Delta \bar{n}_h \leq 0$ signed by Proposition 3. For (65), at fixed $\bar{n}_h$ the set inclusion above raises $\bar{n}_h - |\mathcal{K}_h|$ weakly, so the domestic share weakly rises; the chain-shortening channel removes only stages that are counted in both numerator and denominator of $|\mathcal{K}_h|/\bar{n}_h$ and cannot reverse this ordering. The formal argument is in Appendix A.8. $\square$

Proposition 4 is the theoretical counterpart of the empirical finding that measures of domestic production linkage also decline with participation. Because the two terms in (64) enter with opposite signs, the level n_h^D carries no unconditional prediction: an observed decline in the domestic component identifies the chain-shortening channel as dominant, whereas an increase would identify the retention of formerly imported stages as dominant. The ambiguity is confined to the level; the domestic share rises unambiguously by (65), and it is this share, not the stage count, that the model restricts. The domestic-linkage regression is therefore informative about which of the two theoretical channels prevails, rather than serving as a test of the model's sign predictions.

Corollary 0.4 (Trade-cost moderation of the stage composition). *Suppose Conditions 2 and 1 hold, and let $N'_h > N_h$ and $\tau'_h > \tau_h$, all allocations being evaluated outside Θ^c . Then the contraction of internationally fragmented stages is weakly stronger at higher trade costs,*

$$n_h^G(N'_h, \tau'_h) \leq n_h^G(N'_h, \tau_h) \leq n_h^G(N_h, \tau_h), \quad \frac{\partial^2 n_h^G}{\partial N_h \partial \ln \tau_h} \leq 0, \quad (66)$$

and the same ordering holds for the cross-border stage share $|\mathcal{K}_h|/\bar{n}_h$. The moderation of

the domestic stage count is, by contrast, undetermined: differentiating (64) gives

$$\frac{\partial^2 n_h^D}{\partial N_h \partial \ln \tau_h} = \underbrace{\frac{\partial^2 \bar{n}_h}{\partial N_h \partial \ln \tau_h}}_{\leq 0 \text{ by Corollary 0.3}} - \underbrace{\frac{\partial^2 n_h^G}{\partial N_h \partial \ln \tau_h}}_{\leq 0 \text{ by (66)}}, \quad (67)$$

a difference of two weakly negative terms. Absent Condition 2 the first inequality in (66) holds with equality; amplification along the sourcing margin is therefore a conditional prediction, as is the chain-length amplification of Corollary 0.3.

Sketch. Both channels behind (63) are moderated by τ_h . On the sourcing channel, a higher τ_h raises $\tau_h P_{f,n}/P_{h,n}$ at every stage, so under Condition 2, $\partial^2 \ln(P_{h,n}/\tau_h P_{f,n}) / \partial \ln \omega_h \partial \ln \tau_h \leq 0$, a given proportional wage decline induces a weakly larger contraction of $\mathcal{K}_h$; this is the same step that signs $\partial \kappa_B / \partial \ln \tau_h > 0$ and $\partial \kappa_A / \partial \ln \tau_h < 0$ in Corollary 0.1. On the chain-length channel, Corollary 0.3 delivers a weakly larger reduction in $\bar{n}_h$ under Condition 1, and since $\mathcal{K}_h \subseteq \{1, \dots, \bar{n}_h - 1\}$ by (13), a larger truncation removes weakly more stages from $\mathcal{K}_h$. The two channels reinforce each other, giving (66); the share statement follows because the numerator falls weakly more while the denominator falls, and the ordering of Proposition 4 is preserved. Equation (67) is then an identity. The formal argument is in Appendix A.9. $\square$

2.4.6 Testable Implications

Propositions 1–4 deliver four sets of predictions that map directly into the specifications of Section 3. First, the coefficient on the participation rate is negative for backward participation and positive for forward participation (Proposition 1). Second, it is positive for the GVC position index (Proposition 2). Third, it is negative for measures of chain length and of the internationally fragmented component (Propositions 3 and 4), and negative for the cross-border stage share by (65). Fourth, the interaction of participation with an *inverse* proxy for trade costs, such as the depth of regional trade agreements, carries the sign opposite to the corresponding main effect, by Corollaries 0.1, 0.2, 0.3 and 0.4.

Two qualifications govern how these predictions should be read. The first concerns the interaction terms. The main effects of Propositions 1–4 hold unconditionally, whereas the moderation results are conditional: the sourcing-margin interactions require Condition 2 and the chain-length interaction requires Condition 1, and absent these conditions the relevant cross partials vanish rather than change sign. A statistically insignificant interaction is therefore consistent with the model and is informative about whether the complementarity conditions hold, while an interaction of the *same* sign as the main effect would be evidence against it. Within this set, only the cross partial for the position index is signed strictly; those for the two participation measures and for the integer margins are weak inequalities, so their point estimates are expected to be smaller in absolute value and less precisely signed.

The second qualification concerns the discrete margins. The coefficient on the domestic component is left unsigned by (64), because the chain-shortening and reduced-fragmentation channels enter with opposite signs; its estimate is accordingly interpreted as identifying which of the two channels dominates rather than as a test of a sign prediction. The corresponding prediction with a determinate sign is the one for the cross-border share in (65). Moreover, because Propositions 3 and 4 concern integer margins that are locally constant on $\Theta \setminus \Theta^c$, the associated empirical relationships should be read as holding across the cross section of countries and sectors that differ in their distance from the switching boundaries, rather than as smooth within-unit responses. The same reading applies to the interaction terms for those margins.

3 Empirical Results

Guided by the theory developed in Section 2, we examine how labor supply shapes the organization and vertical structure of global production networks. Specifically, we estimate the effects of the labor force participation rate (LFPR) on the direction of GVC

specialization, the allocation of production stages, and the length of production chains. We also investigate whether these effects vary with international trade costs. Our analysis uses an annual country-sector panel covering 53 economies and 35 sectors over the period 2007–2021.³

3.1 Empirical Specification and Data

We first describe the empirical specifications used to test the theoretical propositions developed in Section 2. We then discuss the data sources, variable construction, and descriptive statistics.

Before turning to causal identification, we examine the partial correlations between labor force participation and dimensions of global value chain integration. Figure 1 displays binned scatter plots relating the labor force participation rate to backward GVC participation, forward GVC participation, the GVC position index, and production linkage yield. All variables are purged of country and year fixed effects to isolate within-country time variation. The evidence indicates that greater labor force participation is associated with longer upstream forward linkages and an upstream shift in GVC position, whereas it correlates with lower backward integration and shorter production linkage yield. We rely on an instrumental-variable strategy in the following section to establish causality.

3.1.1 Regression Specifications

To examine how LFPR affects GVC organization and vertical structure, we estimate the following country-sector-year panel model:

$$GVC_{ict} = \beta_0 + \beta_1 LFPR_{ct} + \beta_2 (LFPR_{ct} \times \ln RTA_{ct}) + \beta_3 \ln RTA_{ct} + \beta_4 X_{ct} + \delta_{ic} + \delta_t + \varepsilon_{ict}. \quad (68)$$

³The empirical sample covers 53 countries: Australia, Austria, Bangladesh, Belgium, Bhutan, Brazil, Bulgaria, Cambodia, Canada, China, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Fiji, Finland, France, Germany, Greece, Hungary, India, Ireland, Italy, Japan, Kazakhstan, Kyrgyzstan, Lao People’s Democratic Republic, Latvia, Lithuania, Luxembourg, Moldova, Mexico, Nepal, Norway, Pakistan, Philippines, Poland, Portugal, Republic of Korea, Russian Federation, Slovakia, Slovenia, Spain, Sri Lanka, Sweden, Switzerland, Thailand, Turkey, United Kingdom, United States, and Viet Nam.

Here, i , c , and t index sectors, countries, and years, respectively. GVC_{ict} denotes a set of outcome variables characterizing the organization and vertical structure of production, including backward GVC participation, forward GVC participation, GVC position, production linkage yield (PLY), average path length (APL), GVC production linkage yield (PLY_GVC), and domestic production linkage yield (PLY_D). These indicators are constructed following the framework of Wang et al. (2013).

The main explanatory variable, $LFPR_{ct}$, is the labor force participation rate in country c and year t . The variable $\ln RTA_{ct}$ measures the depth of regional trade agreements and serves as an inverse proxy for bilateral iceberg trade costs. A deeper RTA lowers trade frictions and therefore corresponds to lower effective trade costs. The interaction term $LFPR_{ct} \times \ln RTA_{ct}$ captures whether trade integration moderates the effect of labor supply on GVC outcomes. The vector X_{ct} contains time-varying country-level controls, including inward FDI and market potential. The term δ_{ic} denotes country-sector fixed effects, while δ_t denotes year fixed effects. Standard errors are clustered at the country-sector level.

A central empirical challenge is the potential endogeneity of LFPR. Reverse causality may arise because GVC expansion and changes in production-chain organization can affect labor demand and draw additional workers into formal employment. Omitted-variable bias may also arise if institutional quality, human capital, demographic structure, or other country-specific factors jointly affect LFPR and GVC organization.

To address these concerns, we instrument LFPR using extreme temperature shocks. The instrument is motivated by evidence that extreme heat and compound hot-dry events can reduce effective labor supply through several channels, including lower agricultural productivity, heat-induced reductions in worker productivity, and constraints on labor-market adaptation (Liu et al., 2023; Somanathan et al., 2021; Parsons et al., 2021; Graff Zivin and Neidell, 2014). The identifying assumption is that, conditional on country-sector fixed ef-

fects, year fixed effects, and the included controls, extreme temperature shocks affect GVC outcomes primarily through their effect on labor supply.

For specifications containing the interaction term, we instrument both endogenous regressors, $LFPR_{ct}$ and $LFPR_{ct} \times \ln RTA_{ct}$, using the temperature-shock instrument and its interaction with $\ln RTA_{ct}$. This approach allows the moderating effect of trade costs to be estimated within a consistent instrumental-variables framework.

3.1.2 Data Sources and Variable Construction

GVC Indicators. Our core GVC indicators are obtained from the Scholar Version of the UIBE GVC Database.⁴ We obtain GVC participation measures from the Index 1 Production Decomposition module. Measures of production length, upstreamness, and production linkage yield are obtained from the Index 3 Production Length module.

LFPR and Instrumental Variable. Country-level monthly LFPR data for the export panel are compiled from the International Labour Organization (ILO), which covers 46 countries over 2017–2021, and the CEIC Global Database, which covers 63 major economies. For economies reporting LFPR only at a quarterly frequency, we assign the quarterly observation to each of the three months in the corresponding quarter. This procedure is commonly used when monthly labor-market data are unavailable.⁵ Annual LFPR data for the GVC panel are obtained from the ILO and the World Bank World Development Indicators (WDI).

The instrumental variable is constructed using extreme temperature data from the National Earth System Science Data Center of China.⁶ Specifically, we use the annual average number of hot-dry days, denoted by *hotdrymeandays*. A hot day is defined as

⁴RIGVC UIBE, 2022, UIBE GVC Database. Available at: <http://gvcd.db.uibe.edu.cn>. The database constructs GVC indicators using the ADB MRIO 2022 release and covers 53 economies and 35 sectors over 2007–2021.

⁵For example, the Q1 value is assigned to January, February, and March.

⁶Available at: <http://www.geodata.cn>. The dataset reports the frequency, duration, and severity of global compound extreme climate events at a spatial resolution of 0.5° for 1951–2022.

a day on which daily maximum temperature exceeds the 95th percentile of the local historical distribution. A dry day is defined as a day on which daily precipitation falls below the 5th percentile of the local historical distribution, calculated using days with positive precipitation. The variable *hotdrymeandays* is the annual average number of days satisfying both criteria.

Trade Cost Variable. For the annual GVC panel, we proxy bilateral trade costs using RTA depth from the DESTA database (Dür et al., 2014). RTA depth is measured as the sum of binary indicators for the substantive provisions covered by the agreement, including FTA status, technical standards, services trade, investment protection, government procurement, competition policy, and intellectual property protection. A higher value indicates deeper regional integration and lower bilateral trade frictions.

Control Variables. Additional time-varying country-level controls include real GDP per capita, household consumption expenditure, population, inward FDI, and total labor-force size. These variables are obtained from the World Bank WDI and the CEIC Global Database. Bilateral geographic distance is obtained from the CEPII Gravity Database. Following Harris (1954), we construct market potential as

$$MP_c = \sum_{k \neq c} \frac{GDP_k}{d_{ck}}, \quad (69)$$

where GDP_k denotes the GDP of country k and d_{ck} denotes the bilateral geographic distance between countries c and k .

Table 1 reports summary statistics for all variables used in the empirical analysis, including the GVC outcomes, production-linkage measures, labor-supply variables, trade-cost measures, and macroeconomic controls.

3.2 Benchmark Regressions

3.2.1 LFPR and GVC Participation

Table 2 reports the baseline 2SLS estimates for backward GVC participation, forward GVC participation, and GVC position. Columns (1), (3), and (5) present specifications without the additional controls, whereas Columns (2), (4), and (6) include the full set of controls reported in the table notes.

The estimates reveal a systematic reorganization of GVC participation as labor supply changes. For backward GVC participation, the coefficient on LFPR in Column (1) is -1.017 and statistically significant at the 1% level. Thus, a one-unit increase in LFPR is associated with a 1.017-unit decline in backward GVC participation. For forward GVC participation, the coefficient in Column (3) is 2.906, indicating that higher LFPR is associated with greater forward participation. The coefficient on GVC position in Column (5) is 0.902, suggesting that higher LFPR shifts production activity toward more upstream positions in the value chain.

These results are consistent with the model’s prediction that greater labor supply lowers domestic labor costs, encourages domestic sourcing, reduces reliance on imported intermediates, and increases the supply of domestic intermediates to downstream producers. The estimates remain qualitatively similar after the inclusion of the full set of controls in Columns (2), (4), and (6). The corresponding 2SLS specifications exhibit strong first-stage relevance, with the reported Wald F-statistics exceeding conventional weak-instrument thresholds.

To assess the economic magnitude of these effects, we follow Korovkin and Makarin (2023) and calculate

$$\Delta \text{GVC Explained} = 100 \times \frac{(LFPR_{p75} - LFPR_{p25}) \left| \widehat{\beta}_1 \right|}{GVC_{p75} - GVC_{p25}}, \quad (70)$$

where $LFPR_{p75}$ and $LFPR_{p25}$ denote the 75th and 25th percentiles of LFPR, while

GVC_{p75} and GVC_{p25} denote the corresponding percentiles of the relevant GVC outcome. The statistic measures the share of the interquartile variation in a GVC outcome that can be accounted for by moving LFPR from its 25th to its 75th percentile.

As reported in Table 3, the interquartile variation in LFPR accounts for 46% of the interquartile variation in backward GVC participation, 62% of the variation in forward GVC participation, and 53% of the variation in GVC position. These magnitudes suggest that labor supply is economically relevant not only for the intensity of GVC participation, but also for its direction and vertical organization.

3.2.2 Trade-Cost Moderation

We next examine whether trade costs moderate the relationship between LFPR and GVC organization. We use RTA depth as an inverse proxy for bilateral trade costs. In Column (1) of Panel A of Table 4, the dependent variable is backward GVC participation. The coefficient on LFPR is negative, whereas the coefficient on the interaction term $LFPR \times \ln RTA$ is positive and statistically significant. Hence, deeper RTAs weaken the negative association between LFPR and backward participation. Equivalently, the negative effect of LFPR is stronger when trade costs are higher.

Column (2) reports the results for forward GVC participation. The coefficient on LFPR is positive, while the interaction coefficient is negative and statistically significant. This pattern indicates that deeper RTAs attenuate the positive effect of LFPR on forward participation. The results for GVC position in Column (3) show a positive baseline effect of LFPR and a negative interaction effect. Thus, higher trade costs strengthen the tendency of greater labor supply to shift production upstream. Overall, these findings are consistent with Propositions 1 and 2.

3.2.3 LFPR and Production-Chain Length

Table 5 reports the 2SLS estimates for four indicators of production-chain structure: *PLY*, *APL*, *PLY_GVC*, and *PLY_D*. The LFPR coefficients are negative and statistically significant at the 1% level in all four specifications. The estimates imply that a one-unit increase in LFPR is associated with a 5.498-unit decline in *PLY*, a 2.913-unit decline in *APL*, a 4.799-unit decline in *PLY_GVC*, and a 3.080-unit decline in *PLY_D*.

The results are consistent with Propositions 3 and 4. In particular, higher LFPR appears to shorten the overall production chain and the internationally fragmented component of production. The negative coefficient on *PLY_D* suggests that the chain-shortening effect dominates the domestic-substitution effect in the aggregate domestic production-linkage measure. This interpretation is consistent with the model, which predicts that higher labor supply can induce firms to substitute domestic for foreign intermediates while simultaneously reducing the total number of production stages.

The economic magnitudes reported in Table 6 are also substantial. Moving LFPR from its 25th to its 75th percentile accounts for 73% of the interquartile variation in *PLY_GVC*, 98.9% of the variation in *APL*, 57% of the variation in *PLY*, and 52% of the variation in *PLY_D*. These estimates indicate that labor supply is closely related to the depth and fragmentation of production networks.

3.2.4 Trade-Cost Moderation of Production-Chain Effects

Table 7 reports the moderating effect of trade costs on production-linkage outcomes. Across all four specifications, the coefficient on LFPR remains negative, consistent with the estimates in Table 5. The interaction coefficient between LFPR and $\ln RTA$ is positive in every regression.

This pattern is consistent with Propositions 3 and 4. Because $\ln RTA$ is an inverse proxy for trade costs, the positive interaction coefficients imply that deeper trade agreements mitigate the negative effect of LFPR on production-chain length and linkage

intensity. Conversely, when trade costs are higher, the decline in domestic wages associated with greater labor supply has stronger implications for input-sourcing decisions, thereby amplifying the contraction of production-chain length. Taken together, the results suggest that labor supply and trade costs jointly shape the vertical architecture of global production networks.

4 Heterogeneity in Industry-level Labor Intensity

Industries differ substantially in their reliance on labor inputs. Our theoretical framework therefore predicts that the effects of LFPR on GVC organization and production-chain structure should be stronger in labor-intensive industries. To test this prediction, we classify sectors according to the median share of labor compensation in value added. Industries with a labor share above the median are defined as labor-intensive, whereas those below the median are classified as less labor-intensive. A higher labor share implies greater reliance on labor inputs and, consequently, greater sensitivity to changes in labor supply and labor costs.

The high-labor-share group includes Agriculture, hunting, forestry and fishing; Food, beverages and tobacco; Textiles and textile products; Leather, leather products and footwear; Wood and products of wood and cork; Pulp, paper, paper products, printing and publishing; Chemicals and chemical products; Rubber and plastics; Basic metals and fabricated metal; Machinery; Electrical and optical equipment; Transport equipment; Manufacturing; Construction; Sale, maintenance and repair of motor vehicles; Wholesale trade and commission trade; Retail trade; Renting of machinery and equipment; and Computer and related activities. The low-labor-share group includes Mining and quarrying; Coke, refined petroleum and nuclear fuel; Other nonmetallic minerals; Electricity, gas and water supply; Hotels and restaurants; Transport and storage; Post and telecommunications; Financial intermediation; Real estate activities; Research and development; Other

business activities; Public administration and defence; Education; Health and social work; Other community, social and personal services; and Private households with employed persons.

Tables 8, 9, and 10 report the heterogeneity results for backward GVC participation, forward GVC participation, and GVC position, respectively. The results are consistent with the theoretical prediction that labor supply matters more in sectors that rely intensively on labor inputs. In the high-labor-share subsample, the coefficient on LFPR is more negative for backward GVC participation and more positive for both forward GVC participation and GVC position. The corresponding estimates are also generally more precisely estimated than those for the low-labor-share subsample. These patterns remain qualitatively similar after controlling for RTA depth, real GDP per capita, and market potential.⁷

These findings suggest that labor supply affects not only the overall intensity of GVC participation but also the direction of specialization within global production networks. In labor-intensive sectors, an increase in LFPR is more likely to reduce reliance on imported intermediates, increase the supply of domestic intermediates, and shift production toward upstream activities. In contrast, sectors with lower labor shares exhibit weaker responses, consistent with their relatively greater reliance on capital, technology, or other non-labor inputs. The estimates for GVC participation and positioning are therefore consistent with the mechanism developed in the model.

Tables 11 and 12 report the corresponding results for production-chain indicators. In the high-labor-share subsample, LFPR significantly reduces production linkage yield, average path length, GVC production linkage yield, and domestic production linkage yield. The estimated effects are generally larger in magnitude in labor-intensive industries, whereas the responses in the low-labor-share subsample are weaker and, in several specifi-

⁷The specifications reported in these tables include the logarithm of RTA depth, the logarithm of real GDP per capita, and the logarithm of market potential as additional controls. All regressions include country, sector, and year fixed effects, and standard errors are clustered at the country-sector level.

cations, less precisely estimated. The main patterns remain robust with and without the additional controls.

The results for the two subcomponents of production linkage yield are also consistent with this interpretation. For example, in the specifications with controls, the estimated effect of LFPR on `PLY_GVC` is -4.552 in the labor-intensive subsample and -5.124 in the low-labor-share subsample, while the corresponding estimates for `PLY_D` are -3.724 and -2.209 , respectively. These estimates indicate that the response of production-chain structure may differ across the global and domestic components of production linkages. Accordingly, the evidence supports the broader conclusion that labor intensity is an important source of heterogeneity in the relationship between labor supply and production-network organization, rather than implying that all dimensions respond uniformly across sectors.

Overall, the heterogeneity results are consistent with the mechanism emphasized in the model. Because labor-intensive industries rely more heavily on labor inputs, changes in labor supply have larger effects on production costs and input-sourcing decisions. These cost changes translate into greater adjustments in GVC participation, GVC positioning, and production-chain length. Industries with lower labor shares, by contrast, exhibit more limited responses along several dimensions. The findings highlight the role of industry factor intensity in shaping how labor supply reorganizes global production networks.

5 Robustness Checks

Sensitivity Analysis. We assess the robustness of the baseline relationship between labor supply and GVC organization by considering alternative specifications, inference procedures, and sample restrictions. Table 15 reports 2SLS estimates using backward GVC participation as the dependent variable.

For ease of comparison, Column (1) reproduces the baseline specification, which

includes the full set of controls, country-sector fixed effects, year fixed effects, and standard errors clustered at the country-sector level. Column (2) excludes all time-varying controls while retaining the fixed effects, thereby assessing whether the estimated LFPR effect is sensitive to the choice of controls. Column (3) instead clusters standard errors at the country level to examine the sensitivity of statistical inference to within-country correlation in the error term. Finally, Column (4) excludes observations from 2020 onward to assess whether the baseline results are driven by the COVID-19 pandemic and its associated disruptions to labor markets and international production networks.

Across all specifications, the coefficient on LFPR remains negative and statistically significant at the 1% level. The point estimate ranges from -1.016 in the specification without controls to -1.433 in the baseline and country-clustered specifications. When the post-2019 observations are excluded, the coefficient remains negative and statistically significant, with an estimated magnitude of -1.213 . These results indicate that the negative relationship between LFPR and backward GVC participation is not sensitive to the inclusion of controls, the level at which standard errors are clustered, or the presence of pandemic-period observations.

The estimates for the control variables also suggest that the pandemic period may have altered the relationship between macroeconomic conditions and GVC participation. In particular, the coefficient on market potential ($\ln mp$) becomes substantially more negative after excluding the COVID-19 period. This pattern should be interpreted cautiously: it suggests that pandemic-era disruptions and policy interventions may have affected the estimated association between market potential and backward GVC participation, rather than establishing a direct effect of monetary policy.

Overall, the sensitivity exercises support the stability of the main empirical pattern: higher LFPR is associated with lower backward GVC participation across a range of reasonable empirical choices. The results therefore appear unlikely to be driven solely by

the baseline control set, the clustering procedure, or pandemic-related observations.

Alternative Instrument Variable. As a robustness check, we implement an alternative instrumental variable strategy that exploits statutory pensionable age reforms to instrument for labor force participation. The instrument captures cross-country and time variation in the legal eligibility age for full pension benefits, compiled from the Social Security Programs Throughout the World database maintained by the U.S. Social Security Administration. To preserve the relevance and validity of this instrument, we exclude Japan, Singapore, and Bangladesh from this exercise. Japan’s exceptionally advanced population ageing implies that changes in the statutory pensionable age may affect GVC outcomes directly through shifts in the age composition of the workforce, rather than solely through aggregate labor force participation. In Singapore, where foreign workers account for a substantial share of total employment, statutory pensionable age reforms primarily apply to resident workers and may therefore have limited explanatory power for aggregate labor force participation. In Bangladesh, the predominance of informal employment substantially weakens the link between statutory retirement provisions and actual labor supply decisions.

The relevance of this instrument is well established in labor economics. Increases in the statutory pensionable age influence individuals’ retirement decisions and working horizons, as many adjust their labor supply to optimize pension entitlements. Indeed, a large body of work robustly confirms a causal effect of statutory pensionable ages on labor force participation (Mohnen, 2025; Mastrobuoni, 2009; Staubli and Zweimüller, 2013; Atalay and Barrett, 2015). First-stage estimates from our sample are consistent with this established finding. The exclusion restriction rests on the premise that changes in statutory pensionable ages affect GVC outcomes primarily through their impact on labor supply, rather than through other channels. Pension reforms are typically driven by broad macroeconomic and demographic pressures such as fiscal sustainability and population

aging, not by targeted industrial policies aimed at shaping GVC integration. Governments do not adjust retirement ages with the explicit goal of altering GVC participation in specific sectors. To the extent that confounding macro factors correlate with pension reforms, they are absorbed by the set of country fixed effects, year fixed effects, and time-varying controls included in the specification.

Results from this robustness check, reported in Appendix Table 16, are broadly consistent with our baseline findings. In line with the main estimates in Table 2, we find that labor force participation is negatively associated with backward GVC participation, and positively associated with forward GVC participation and the GVC position index. Echoing the patterns in Table 5, labor force participation continues to enter negatively and significantly for Production Linkage Yield, Average Path Length, GVC-related Production Linkage Yield, and Domestic Production Linkage Yield. The consistent estimates across all GVC indicators confirm that our core conclusions are robust to the choice of instrumental variable.

6 Conclusion

This paper examines how labor supply shapes the organization and vertical structure of global value chains. Motivated by persistent changes in labor-force participation associated with population aging, climate change, pandemics, and geopolitical disruptions, we develop a general-equilibrium model of endogenous multi-stage production. In the model, labor supply affects equilibrium wages, input-sourcing decisions, and the length of production chains. A higher labor-force participation rate (LFPR) expands domestic labor supply, lowers equilibrium wages, and reduces the cost of domestically produced intermediates. Firms therefore substitute domestic sourcing for imported inputs. As a result, higher LFPR reduces backward GVC participation, increases forward GVC participation, shifts production toward more upstream positions, and shortens internationally

fragmented production chains. These effects are stronger when cross-border trade costs are higher.

Using an annual country–sector panel covering 53 economies and 35 sectors over 2007–2021, we find empirical evidence consistent with these theoretical predictions. We employ an instrumental-variable strategy based on compound extreme-temperature shocks to address the potential endogeneity of LFPR. The results should therefore be interpreted under the assumption that, conditional on the fixed effects and controls, these shocks affect GVC outcomes primarily through their effects on labor supply. A one-percentage-point increase in LFPR is associated with a 1.43-percentage-point reduction in backward GVC participation, a 2.54-percentage-point increase in forward GVC participation, and a 2.91-unit reduction in average production length. Higher LFPR is also associated with a more upstream GVC position. These findings indicate that labor supply affects not only the intensity of GVC participation, but also the direction of specialization and the vertical organization of international production.

The heterogeneity analysis shows that these relationships are generally stronger in labor-intensive industries, where labor costs account for a larger share of value added. In such sectors, changes in labor supply generate larger adjustments in sourcing decisions, GVC participation, GVC positioning, and production-chain structure. At the same time, the effects are not identical across all production-linkage measures. This pattern suggests that labor supply may influence the domestic and international components of production fragmentation through partly distinct channels.

Trade integration provides an important qualification to these results. Deeper regional trade agreements, which we use as an inverse proxy for bilateral trade costs, generally attenuate the relationship between LFPR and GVC participation and production-chain outcomes. Lower trade barriers make cross-border production sharing less costly and thereby weaken the incentive to substitute domestic for foreign sourcing. Labor supply

and trade integration should therefore be viewed as interconnected determinants of GVC organization rather than as independent forces.

Taken together, the findings identify labor supply as an important and underexplored determinant of GVC organization. The existing literature has emphasized technology, comparative advantage, contractual frictions, trade costs, and institutions as major forces shaping global production. Our results complement this literature by showing that cross-country differences in labor-supply conditions can reorganize the direction of specialization, the allocation of production stages, and the extent of international production fragmentation. More broadly, demographic and labor-market changes may have persistent implications for the geography and vertical structure of global value chains.

Several avenues remain for future research. Distinguishing between skilled and unskilled labor participation could clarify whether different types of labor affect sourcing and production organization through distinct mechanisms. Future work could also examine how labor-market institutions, migration, and labor quality interact with LFPR in shaping GVC restructuring. Finally, firm-level and worker-level data could provide further evidence on the mechanisms linking labor supply to wages, domestic sourcing, and the international reallocation of production stages.

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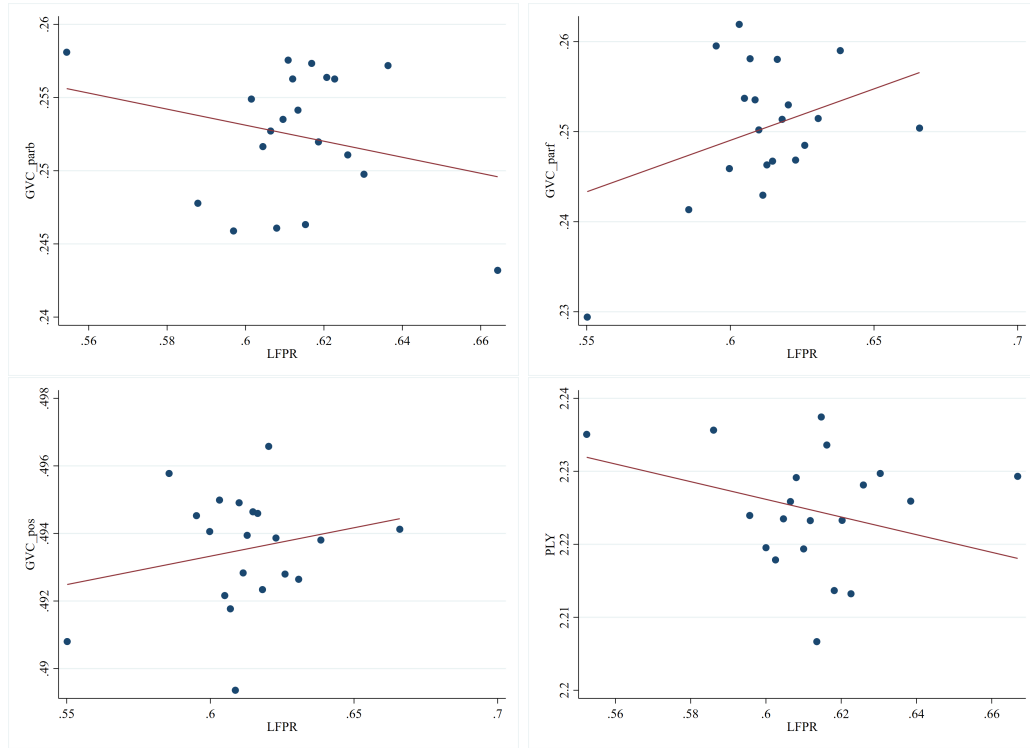
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Figure 1: GVC Participation, Position and production linkage yield V.S. Labor Force Participation Rate



Note: This figure presents binned scatter plots showing the conditional correlations between four global value chain (GVC) measures and the labor force participation rate (LFPR). GVC_parb denotes backward GVC participation; GVC_parf denotes forward GVC participation; GVC_pos is the GVC position index; and PLY represents production linkage yield. All variables are residualized after absorbing country and year fixed effects. The solid lines depict the linear fitted relationships.

Table 1: Summary Statistics of Key Variables

Variable	Mean	SD	Observations	Min	Max
Dependent Variables					
GVC_parb	0.245794	0.163426	24,487	0.000000	1.187137
GVC_parf	0.246184	0.299493	26,004	0.000000	21.583105
GVC_pos	0.494886	0.149029	26,530	0.168140	1.000000
PLY	2.211576	0.478067	25,530	1.000000	5.947412
APL	1.666704	0.171982	25,530	1.011853	5.204391
PLY_D	1.659587	0.330101	25,530	0.325461	5.282080
PLY_GVC	4.054929	0.383370	25,530	2.641762	7.518580
Control Variables					
LFPR	0.612755	0.066906	26,582	0.393000	0.854400
Indisas	2.478700	0.553187	27,175	0.757686	4.166665
lnRTA	3.854517	1.888504	27,296	0.000000	5.710427
lnpgdp	9.606150	1.364734	29,247	5.957002	12.318620
lnmp	22.130820	6.158290	26,809	0.000000	25.008230

Notes: [1] Dependent Variables: GVC_parb = backward GVC participation; GVC_parf = forward GVC participation; GVC_pos = GVC position index; PLY = production linkage yield; APL = average production linkage yield; PLY_D = domestic production linkage yield; PLY_GVC = GVC-related production linkage yield. [2] Control Variables: LFPR = labor force participation rate; Indisas = log disaster intensity (instrumental variable); lnRTA = log regional trade agreements; lnpgdp = log GDP per capita; lnmp = log market potential. [3] SD = Standard Deviation; all log-transformations use natural logarithm (ln). [4] Data sources: Dependent Variables – UIBE GVC Index; Control variables – World Bank WDI, IMF, and national statistical databases; Instrumental variable – National Earth System Science Data Center. [5] The empirical sample covers 53 countries: Australia, Austria, Bangladesh, Belgium, Bhutan, Brazil, Bulgaria, Cambodia, Canada, China, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Fiji, Finland, France, Germany, Greece, Hungary, India, Ireland, Italy, Japan, Kazakhstan, Kyrgyzstan, Lao People’s Democratic Republic, Latvia, Lithuania, Luxembourg, Moldova, Mexico, Nepal, Norway, Pakistan, Philippines, Poland, Portugal, Republic of Korea, Russian Federation, Slovakia, Slovenia, Spain, Sri Lanka, Sweden, Switzerland, Thailand, Turkey, United Kingdom, United States, and Viet Nam.

Table 2: 2SLS Regression Results for GVC Participation and Position

Panel A: Second Stage Regressions						
	GVC_parb		GVC_parf		GVC_pos	
	(1)	(2)	(3)	(4)	(5)	(6)
LFPR	-1.017*** (0.357)	-1.433*** (0.399)	2.906*** (0.965)	2.541** (1.092)	0.902*** (0.200)	1.172*** (0.237)
Control Variables	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24,483	22,905	25,999	24,422	26,530	24,920
Panel B: First Stage Regressions (Dep. Var = LFPR)						
	GVC_parb		GVC_parf		GVC_pos	
	(1)	(2)	(3)	(4)	(5)	(6)
IV	-0.003*** (0.000)	-0.003*** (0.000)	-0.004*** (0.000)	-0.003*** (0.000)	-0.004*** (0.000)	-0.003*** (0.000)
Control Variables	No	Yes	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Wald F Statistic	47.20	43.54	91.10	72.44	101.3	80.13
LM Statistic	50.49	46.93	96.25	77.33	106.8	85.46
Observations	24,483	22,905	25,999	24,422	26,530	24,920

Notes: [1] This table presents the baseline results of Two-Stage Least Squares (2SLS) regressions. Panel A reports the second-stage estimates, where the dependent variables are GVC backward participation (GVC_parb, Columns 1–2), GVC forward participation (GVC_parf, Columns 3–4), and GVC position (GVC_pos, Columns 5–6). Panel B reports the first-stage regressions. [2] Dependent variable is the labor force participation rate (LFPR). The instrumental variable (IV) is hot-dry mean days ('hotdrymeandays'), sourced from the National Earth System Science Data Center, National Science & Technology Infrastructure of China (<http://www.geodata.cn>). Columns (2), (4), and (6) include the following control variables: the logarithm of the Regional Trade Agreement (RTA) depth index, the logarithm of real GDP per capita, and the logarithm of market potential. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are reported in parentheses. [4] The Wald F statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [5] All regressions include country, sector, and year fixed effects to absorb unobserved time-invariant country/sector heterogeneity and time-varying global common shocks.

Table 3: Explanatory Power of LFPR on GVC Participation and Position

Variables	P_{25}	P_{75}	$P_{75}-P_{25}$	$\hat{\beta}_1$	Change explained
LFPR	0.571	0.644	0.072		
GVC_parb	0.118	0.344	0.226	-1.433	45.653%
GVC_parf	0.073	0.366	0.293	2.541	62.441%
GVC_pos	0.394	0.552	0.158	1.172	53.408%

Notes: [1] This table quantifies the explanatory power of the labor force participation rate (LFPR) on GVC indicators using quantile differences ($P_{75} - P_{25}$). P_{25} and P_{75} denote the 25th and 75th percentiles (1st and 3rd quartiles) of the respective variables. [2] The explanatory power (% of Change Explained) in the final column is calculated as $(|\hat{\beta}_1| \times (P_{75} - P_{25})_{\text{LFPR}}) / (P_{75} - P_{25})_{\text{GVC indicator}} \times 100\%$. [3] $\hat{\beta}_1$ estimates for GVC_parb, GVC_parf, and GVC_pos are obtained from Columns (2), (4), and (6) of Table 2, respectively (LFPR has no β as the explanatory variable). [4] LFPR data are sourced from the World Bank; all other data are obtained from the UIBE GVC Database.

Table 4: Trade Cost Moderation on GVC Participation and Position

Panel A: Second Stage Regressions			
	GVC_parb	GVC_parf	GVC_pos
LFPR	-0.687*** (0.154)	0.856*** (0.201)	0.374*** (0.069)
LFPR $\times$ lnRTA	0.240*** (0.075)	-0.426* (0.233)	-0.213*** (0.040)
lnRTA	-0.131*** (0.042)	0.277* (0.142)	0.126*** (0.024)
Control Variables	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	22905	24422	24920
Panel B: First Stage Regressions (Dep. Var = LFPR)			
	GVC_parb	GVC_parf	GVC_pos
IV	-0.047*** (0.002)	-0.045*** (0.002)	-0.045*** (0.002)
IV $\times$ lnRTA	0.010*** (0.0005)	0.010*** (0.0004)	0.010*** (0.0004)
lnRTA	-0.025*** (0.002)	-0.025*** (0.001)	-0.025*** (0.001)
Control Variables	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Wald F Statistic	113.555	182.860	190.286
LM Statistic	189.329	265.464	275.760
Observations	22905	24422	24920

Notes: [1] This table presents the results of Two-Stage Least Squares (2SLS) regressions with interaction effects. Panel A reports the second-stage estimates, where the dependent variables are GVC backward participation (GVC_parb), GVC forward participation (GVC_parf), and GVC position (GVC_pos). Panel B reports the first-stage regressions, with the dependent variable being the labor force participation rate (LFPR). [2] The instrumental variable (IV) is hot-dry mean days ('hotdrymeandays'). All specifications include the following control variables: the logarithm of real GDP per capita and the logarithm of market potential. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses. [4] All regressions include country, sector, and year fixed effects to absorb unobserved time-invariant country/sector heterogeneity and time-varying global common shocks.

Table 5: 2SLS Regression Results for GVC Production Length Indicators

Panel A: Second Stage Regressions				
Dependent Variable	(1) PLY	(2) APL	(3) PLY_GVC	(4) PLY_D
LFPR	-5.498*** (1.028)	-2.913*** (0.518)	-4.799*** (0.970)	-3.080*** (0.705)
Control Variables	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	23,985	23,985	23,985	23,985
Panel B: First Stage Regressions (Dep. Var = LFPR)				
	(1) PLY	(2) APL	(3) PLY_GVC	(4) PLY_D
IV	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
Control Variables	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Wald F Statistic	72.66	72.66	72.66	72.66
LM Statistic	77.56	77.56	77.56	77.56
Observations	23,985	23,985	23,985	23,985

Notes: [1] This table presents the results of Two-Stage Least Squares (2SLS) regressions. Panel A reports the second-stage estimates, where the dependent variables are core indicators of GVC production length: (1) PLY (production linkage yield), (2) APL (average path length), (3) PLY_GVC (GVC-related production linkage yield), and (4) PLY_D (domestic production linkage yield). Panel B reports the first-stage regressions. [2] Variable definitions and specification notes: LFPR = labor force participation rate. The instrumental variable (IV) is hot-dry mean days ('hotdrymeandays'), sourced from the National Earth System Science Data Center, National Science & Technology Infrastructure of China (<http://www.geodata.cn>). All specifications include the following control variables: the logarithm of the Regional Trade Agreement (RTA) depth index, the logarithm of real GDP per capita, and the logarithm of market potential. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses. [4] The Wald F-statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [5] All regressions include country, sector, and year fixed effects to absorb unobserved time-invariant country/sector heterogeneity and time-varying global common shocks.

Table 6: Explanatory Power of LFPR on GVC Production Length

Variables	P_{25}	P_{75}	$P_{75}-P_{25}$	$\hat{\beta}_1$	Change explained
LFPR	0.571	0.644	0.072		
PLY	1.862	2.551	0.689	-5.498	57.453%
APL	1.551	1.763	0.212	-2.913	98.932%
PLY_GVC	3.812	4.284	0.472	-4.799	73.205%
PLY_D	1.427	1.852	0.425	-3.080	52.179%

Notes: [1] This table quantifies the explanatory power of the labor force participation rate (LFPR) on GVC production length indicators using quantile differences ($P_{75} - P_{25}$). P_{25} and P_{75} denote the 25th and 75th percentiles (1st and 3rd quartiles) of the respective variables. [2] Explanatory power (% of Change Explained) is calculated as: $(|\hat{\beta}_1| \times (P_{75} - P_{25})_{\text{LFPR}}) / (P_{75} - P_{25})_{\text{GVC indicator}} \times 100\%$. [3] $\hat{\beta}_1$ estimates for PLY, APL, PLY_GVC, and PLY_D are obtained from Columns (1)–(4) of Table 5, respectively (LFPR has no β as the explanatory variable). [4] Data sources: LFPR (World Bank); PLY/APL/PLY_GVC/PLY_D (UIBE GVC Database).

Table 7: Trade Cost Moderation of LFPR on GVC Length

Panel A: Second Stage Regressions				
	PLY	APL	PLY_GVC	PLY_D
LFPR	-1.772*** (0.242)	-0.879*** (0.116)	-1.103*** (0.232)	-1.147*** (0.190)
LFPR×lnRTA	0.948*** (0.155)	0.518*** (0.074)	0.941*** (0.149)	0.492*** (0.122)
lnRTA	-0.55093*** (0.09347)	-0.30426*** (0.04480)	-0.53904*** (0.08977)	-0.28895*** (0.07358)
Control Variables	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	23985	23985	23985	23985
Panel B: First Stage Regressions (Dep. Var = LFPR)				
	PLY	APL	PLY_GVC	PLY_D
IV	-0.045*** (0.001)	-0.045*** (0.001)	-0.045*** (0.001)	-0.045*** (0.001)
IV×lnRTA	0.010*** (0.0002)	0.010*** (0.0002)	0.010*** (0.0002)	0.10*** (0.0002)
lnRTA	-0.025*** (0.0007)	-0.025*** (0.0007)	-0.025*** (0.0007)	-0.025*** (0.0007)
Control Variables	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Wald F Statistic	136.002	136.002	136.002	136.002
LM Statistic	290.150	290.150	290.150	290.150
Observations	23985	23985	23985	23985

Notes: [1] This table presents Two-Stage Least Squares (2SLS) regression results with interaction effects. Panel A reports second-stage estimates, where the dependent variables are core indicators of GVC production length: (1) PLY (production linkage yield), (2) APL (average path length), (3) PLY_GVC (GVC-related production linkage yield), and (4) PLY_D (domestic production linkage yield). Panel B reports first-stage regressions, with the dependent variable being the labor force participation rate (LFPR). [2] The instrumental variable (IV) is hot-dry mean days ('hotdrymeandays'). All specifications include the following control variables: the logarithm of real GDP per capita and the logarithm of market potential. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses. [4] All regressions include country, sector, and year fixed effects to absorb unobserved time-invariant country/sector heterogeneity and time-varying global common shocks.

Table 8: Regression Results for GVC Backward Participation by Labor Sensitivity

Dependent Variable: GVC_parb	(1)	(2)	(3)	(4)	(5)	(6)
Sample	Full Sample	Labor-Sensitive	Non-Labor-Sensitive	Full Sample	Labor-Sensitive	Non-Labor-Sensitive
LFPR	-1.017*** (0.357)	-1.103** (0.446)	-0.901 (0.586)	-1.433*** (0.399)	-1.663*** (0.520)	-1.125* (0.628)
lnRTA				0.0105*** (0.00292)	0.0174*** (0.00406)	0.00195 (0.00420)
Control Variables	No	No	No	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Wald F Statistic	47.20	27.86	19.37	43.54	25.73	17.82
LM Statistic	50.49	29.81	20.75	46.93	27.75	19.24
Observations	24,483	13,412	11,071	22,905	12,547	10,358

Notes: [1] This table presents instrumental variable (IV) regression results for GVC backward participation (GVC_parb). LFPR = labor force participation rate. Columns (1)–(3) report results for the full sample, labor-sensitive subsample, and non-labor-sensitive subsample without additional control variables, respectively. Columns (4)–(6) present results for the same subsamples with control variables included: lnRTA (log of Regional Trade Agreement depth index), log real GDP per capita, and log market potential. All specifications include country, sector, and year fixed effects. [2] Instrumental variable (IV): hot-dry mean days ('hotdrymeandays'). Wald F-statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses.

Table 9: Regression Results for GVC Forward Participation by Labor Sensitivity

Dependent Variable: GVC_parf	(1)	(2)	(3)	(4)	(5)	(6)
Sample	Full Sample	Labor-Sensitive	Non-Labor-Sensitive	Full Sample	Labor-Sensitive	Non-Labor-Sensitive
LFPR	2.906*** (0.965)	2.006*** (0.493)	4.077* (2.147)	2.541** (1.092)	1.817*** (0.527)	3.484 (2.446)
lnRTA				0.0184*** (0.00309)	0.0196*** (0.00230)	0.0170*** (0.00632)
Control Variables	No	No	No	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Wald F Statistic	91.10	52.69	38.41	72.44	42.09	30.35
LM Statistic	96.25	55.68	40.64	77.33	44.94	32.45
Observations	25,999	14,239	11,760	24,422	13,374	11,048

Notes: [1] This table presents instrumental variable (IV) regression results for GVC forward participation (GVC_parf). LFPR = labor force participation rate. Columns (1)–(3) report results for the full sample, labor-sensitive subsample, and non-labor-sensitive subsample without additional control variables, respectively. Columns (4)–(6) present results for the same subsamples with control variables included: lnRTA (log of Regional Trade Agreement depth index), log real GDP per capita, and log market potential. All specifications include country, sector, and year fixed effects. [2] Instrumental variable (IV): hot-dry mean days ('hotdrymeandays'). Wald F-statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses.

Table 10: Regression Results for GVC Position by Labor Sensitivity

Dependent Variable: GVC_pos	(1)	(2)	(3)	(4)	(5)	(6)
Sample	Full Sample	Labor-Sensitive	Non-Labor-Sensitive	Full Sample	Labor-Sensitive	Non-Labor-Sensitive
LFPR	0.902*** (0.200)	1.044*** (0.264)	0.733** (0.307)	1.172*** (0.237)	1.350*** (0.319)	0.961*** (0.359)
lnRTA				-0.00422*** (0.00107)	-0.00533*** (0.00141)	-0.00292* (0.00166)
Control Variables	No	No	No	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Wald F Statistic	101.3	54.97	46.28	80.13	43.47	36.60
LM Statistic	106.8	58.00	48.84	85.46	46.39	39.07
Observations	26,530	14,402	12,128	24,920	13,528	11,392

Notes: [1] This table presents instrumental variable (IV) regression results for GVC Position (GVC_pos). LFPR = labor force participation rate. Columns (1)–(3) report results for the full sample, labor-sensitive subsample, and non-labor-sensitive subsample without additional control variables, respectively. Columns (4)–(6) present results for the same subsamples with control variables included: lnRTA (log of Regional Trade Agreement depth index), log real GDP per capita, and log market potential. All specifications include country, sector, and year fixed effects. [2] Instrumental variable (IV): hot-dry mean days ('hotdrymeandays'). Wald F-statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses.

Table 11: Regression Results for GVC Production Length (PLY and APL) by Labor Sensitivity

Dependent Variable	PLY			APL		
Sample	(1) Full Sample	(2) Labor-Sensitive	(3) Non-Labor-Sensitive	(4) Full Sample	(5) Labor-Sensitive	(6) Non-Labor-Sensitive
LFPR	-5.498*** (1.028)	-6.113*** (1.409)	-4.680*** (1.502)	-2.913*** (0.518)	-2.735*** (0.686)	-3.140*** (0.790)
lnRTA	0.0256*** (0.00476)	0.0338*** (0.00655)	0.0156** (0.00696)	0.0105*** (0.00256)	0.0122*** (0.00341)	0.00832** (0.00387)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Wald F Statistic	72.66	42.05	30.57	72.66	42.05	30.57
LM Statistic	77.56	44.90	32.68	77.56	44.90	32.68
Observations	23,985	13,359	10,626	23,985	13,359	10,626

Notes: [1] This table presents instrumental variable (IV) regression results for core indicators of GVC production length: PLY (production linkage yield, Columns 1–3) and APL (average path length, Columns 4–6). LFPR = labor force participation rate. Columns (1)–(3) and (4)–(6) report results for the full sample, labor-sensitive subsample, and non-labor-sensitive subsample, respectively. All specifications include control variables: lnRTA (log of Regional Trade Agreement depth index), log real GDP per capita, and log market potential, as well as country, sector, and year fixed effects. [2] Instrumental variable (IV): hot-dry mean days ('hotdrymeandays'). Wald F-statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses.

Table 12: Regression Results for GVC Production Length (PLY_GVC and PLY_D) by Labor Sensitivity

Dependent Variable Sample	PLY_GVC			PLY_D		
	(1) Full Sample	(2) Labor-Sensitive	(3) Non-Labor-Sensitive	(4) Full Sample	(5) Labor-Sensitive	(6) Non-Labor-Sensitive
LFPR	-4.799*** (0.970)	-4.552*** (1.206)	-5.124*** (1.595)	-3.080*** (0.705)	-3.724*** (0.998)	-2.209** (0.983)
lnRTA	0.0330*** (0.00427)	0.0372*** (0.00568)	0.0277*** (0.00642)	0.0101*** (0.00363)	0.0169*** (0.00504)	0.00169 (0.00521)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Wald F Statistic	72.66	42.05	30.57	72.66	42.05	30.57
LM Statistic	77.56	44.90	32.68	77.56	44.90	32.68
Observations	23,985	13,359	10,626	23,985	13,359	10,626

Notes: [1] This table presents instrumental variable (IV) regression results for two sub-indicators of GVC production length: PLY_GVC (GVC production linkage yield, Columns 1–3) and PLY_D (domestic production linkage yield, Columns 4–6). LFPR = labor force participation rate. Columns (1)–(3) and (4)–(6) report results for the full sample, labor-sensitive subsample, and non-labor-sensitive subsample, respectively. All specifications include control variables: ln RTA (log of Regional Trade Agreement depth index), log real GDP per capita, and log market potential, as well as country, sector, and year fixed effects. [2] Instrumental variable (IV): hot-dry mean days ('hotdrymeandays'). Wald F-statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses.

Table 13: Regression Results for GVC Production Length (PLY and APL) by Labor Sensitivity (No Control Variables)

Dependent Variable Sample	PLY			APL		
	(1) Full Sample	(2) Labor-Sensitive	(3) Non-Labor-Sensitive	(4) Full Sample	(5) Labor-Sensitive	(6) Non-Labor-Sensitive
LFPR	-3.686*** (0.851)	-4.110*** (1.106)	-3.141** (1.231)	-1.981*** (0.388)	-1.750*** (0.549)	-2.281*** (0.613)
Wald F Statistic	59.62	52.83	38.70	59.62	52.83	38.70
LM Statistic	61.64	55.84	40.93	61.64	55.84	40.93
Observations	25,530	14,221	11,309	25,530	14,221	11,309

Notes: [1] This table presents instrumental variable (IV) regression results for core indicators of GVC production length: PLY (production linkage yield, Columns 1–3) and APL (average path length, Columns 4–6). LFPR = labor force participation rate. Columns (1)–(3) and (4)–(6) report results for the full sample, labor-sensitive subsample, and non-labor-sensitive subsample, respectively. [2] Instrumental variable (IV): hot-dry mean days ('hotdrymeandays'). Wald F-statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses.

Table 14: Regression Results for GVC Production Length (PLY_GVC and PLY_D) by Labor Sensitivity (No Control Variables)

Dependent Variable Sample	PLY_GVC			PLY_D		
	(1) Full Sample	(2) Labor-Sensitive	(3) Non-Labor-Sensitive	(4) Full Sample	(5) Labor-Sensitive	(6) Non-Labor-Sensitive
LFPR	-2.479*** (0.742)	-2.462*** (0.951)	-2.504** (1.216)	-2.793*** (0.661)	-3.219*** (0.878)	-2.224** (0.912)
Wald F Statistic	59.62	52.83	38.70	59.62	52.83	38.70
LM Statistic	61.64	55.84	40.93	61.64	55.84	40.93
Observations	25,530	14,221	11,309	25,530	14,221	11,309

Notes: [1] This table presents instrumental variable (IV) regression results for two sub-indicators of GVC production length: PLY_GVC (GVC production linkage yield, Columns 1–3) and PLY_D (domestic production linkage yield, Columns 4–6). LFPR = labor force participation rate. Columns (1)–(3) and (4)–(6) report results for the full sample, labor-sensitive subsample, and non-labor-sensitive subsample, respectively. [2] Instrumental variable (IV): hot-dry mean days ('hotdrymeandays'). Wald F-statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [3] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses.

Table 15: Robustness Checks: Sensitivity Analysis

	(1)	(2)	(3)	(4)
GVC_parb	Baseline	No Controls	Country-Cluster	No COVID
LFPR	-1.433*** (0.399)	-1.016*** (0.357)	-1.433*** (0.455)	-1.213*** (0.433)
lnRTA	0.0105*** (0.0029)		0.0105** (0.0042)	0.0121*** (0.0043)
lnpgdp	-0.0272*** (0.0033)		-0.0272*** (0.0047)	0.0197 (0.0157)
lnmp	-0.0340*** (0.0079)		-0.0340*** (0.0094)	-0.4195*** (0.1190)
Country FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	22905	22905	22905	21287

Notes: [1] This table presents 2SLS instrumental variable estimates for the impact of labor force participation rate (LFPR) on GVC participation (GVC_parb). *lndisas* (log disaster intensity) serves as the instrument for LFPR. All regressions include country-sector and year fixed effects, with standard errors reported in parentheses. [2] Column (1): Baseline specification with full controls (*lnRTA*, *lnpgdp*, *lnmp*) and country-sector clustered standard errors. LFPR shows a significant negative effect on GVC participation. [3] Column (2): Specification excluding all control variables. The negative and significant LFPR coefficient remains robust. [4] Column (3): Standard errors clustered at the country level. Estimates are consistent with the baseline results. [5] Column (4): COVID-19 period (2020–2022) observations are excluded. The negative LFPR effect remains significant with a slightly smaller magnitude. Notably, the negative effect of monetary policy (*lnmp*) is substantially strengthened, implying a stronger pandemic-era monetary policy impact on GVC participation. [6] ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 16: Robustness: Alternative Instrument Variable

Panel A: Second Stage Regressions							
	GVC_parb	GVC_parf	GVC_pos	PLY	APL	PLY_GVC	PLY_D
LFPR	-0.315** (-2.03)	1.181*** (2.98)	0.260*** (3.06)	-2.305*** (-5.37)	-0.852*** (-4.94)	-2.136*** (-5.96)	-0.975** (-2.69)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20560	20563	20965	20198	20198	20198	20198
Panel B: First Stage Regressions (Dep. Var = LFPR)							
	GVC_parb	GVC_parf	GVC_pos	PLY	APL	PLY_GVC	PLY_D
IV	0.004*** (12.75)	0.007*** (20)	0.004*** (13.59)	0.004*** (13.58)	0.004*** (13.58)	0.004*** (13.58)	0.004*** (13.58)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wald F Statistic	162.5	400.1	184.8	184.5	184.5	184.5	184.5
LM Statistic	215.3	578.6	243.8	244.9	244.9	244.9	244.9
Observations	20560	20563	20965	20198	20198	20198	20198

Notes: [1] This table reports Two-Stage Least Squares (2SLS) estimates using an alternative instrument. The instrument is the statutory pensionable age for full benefit eligibility, compiled from the *Social Security Programs Throughout the World* database maintained by the U.S. Social Security Administration. [2] The dependent variables, by column, are: (1) GVC backward participation (GVC_parb); (2) GVC forward participation (GVC_parf); (3) GVC position (GVC_pos); (4) production linkage yield (PLY); (5) average path length (APL); (6) GVC-related production linkage yield (PLY_GVC); and (7) domestic production linkage yield (PLY_D). LFPR denotes the labor force participation rate. [3] As discussed in the text, Japan, Singapore, and Bangladesh are excluded from this exercise. [4] All specifications include the logarithm of the Regional Trade Agreement (RTA) depth index, the logarithm of real GDP per capita, and the logarithm of market potential. [5] ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered at the country-sector level and reported in parentheses. [6] The Wald F statistic and LM statistic are reported to test for weak instruments (instrument relevance) and underidentification, respectively. [7] All regressions include country, sector, and year fixed effects to absorb unobserved time-invariant country/sector heterogeneity and time-varying global common shocks.

Appendix For Online Publication

A Proofs of Propositions

Throughout, comparative statics are taken under Assumption 5, so that foreign stage prices $\{P_{f,n}\}_n$ and the foreign wage ω_f are invariant to N_h and ω_h , and are evaluated at parameters outside the critical set Θ^c of Section 2.2.3(iii), where the equilibrium is locally unique and the configuration $d = (\bar{n}_h, \bar{n}_f, \mathcal{K}_h, \mathcal{K}_f)$ is locally constant. Coordination services are the numeraire, $p^M \equiv 1$.

A.1 Preliminaries: elasticities and maintained conditions

Aggregate labor demand.

Lemma 6 (Derived demand for stage output). *Fix $d \in \mathcal{D}$ and let $\zeta_{i,n} \equiv -\partial \ln X_{i,n} / \partial \ln P_{i,n}$ denote the own-price elasticity of demand for the output of stage n of country i , holding wages and foreign prices fixed. Then:*

(i) *At the terminal stage, $\zeta_{i,\bar{n}_i} \geq \min\{\sigma_{X_i}, \sigma\} > 1$.*

(ii) *For $n < \bar{n}_i$ with $n \notin \mathcal{K}_i$ (the stage- n good is used only by the domestic stage $n+1$),*

$$\zeta_{i,n} = \lambda_{i,n+1} + (1 - \lambda_{i,n+1}) s_{i,n+1} \zeta_{i,n+1}, \quad (\text{A.1})$$

with $s_{i,n+1}$ defined in (35). If in addition the stage- n good is exported ($n \in \mathcal{K}_j$), $\zeta_{i,n}$ is a convex combination, with weights equal to the domestic and foreign expenditure shares on $X_{i,n}$, of the right-hand side of (A.1) and of the analogous foreign term $\lambda_{j,n+1} + (1 - \lambda_{j,n+1}) s_{j,n+1} \zeta_{j,n+1}$.

(iii) *Consequently $\zeta_{i,n} \geq \min\{\lambda_{i,n+1}, \lambda_{j,n+1}\} > 0$ for every $n < \bar{n}_i$, and $\zeta_{i,n} > 0$ for every active stage $n \leq \bar{n}_i$.*

Proof. (i) At the terminal stage, CES demand from the aggregator (24), with the share elasticity (53) and $\sigma_{X_i} = 1 + \alpha_i$, and from final consumers (5) with elasticity σ , implies

that expenditure on $X_{i,\bar{n}_i}$ falls by at least $(\min\{\sigma_{X_i}, \sigma\} - 1)$ percent when $P_{i,\bar{n}_i}$ rises by one percent; since quantity equals expenditure divided by price, $\zeta_{i,\bar{n}_i} \geq \min\{\sigma_{X_i}, \sigma\}$. Foreign aggregator demand for an exported terminal good adds a nonnegative term because it is also CES with $\sigma_{X_j} > 1$. (ii) Suppose first that the stage- n good is used only by the domestic stage $n + 1$. By Shephard's lemma applied to (A.8), expenditure on the stage- n input is a constant fraction of downstream variable cost,

$$P_{i,n} X_{i,n} = (1 - \lambda_{i,n+1}) V_{i,n+1} X_{i,n+1},$$

so that

$$\ln X_{i,n} = \ln(1 - \lambda_{i,n+1}) + \ln V_{i,n+1} + \ln X_{i,n+1} - \ln P_{i,n}.$$

From (A.8), $\partial \ln V_{i,n+1} / \partial \ln P_{i,n} = 1 - \lambda_{i,n+1}$, and since $P_{i,n+1} = V_{i,n+1} + \chi^n$ with χ^n independent of $P_{i,n}$, $\partial \ln P_{i,n+1} / \partial \ln P_{i,n} = s_{i,n+1}(1 - \lambda_{i,n+1})$. Hence $\partial \ln X_{i,n+1} / \partial \ln P_{i,n} = -\zeta_{i,n+1} s_{i,n+1}(1 - \lambda_{i,n+1})$, and differentiating the display above,

$$-\zeta_{i,n} = (1 - \lambda_{i,n+1}) - \zeta_{i,n+1} s_{i,n+1}(1 - \lambda_{i,n+1}) - 1,$$

which rearranges to (A.1). If the stage- n good is also exported, total demand is the sum of domestic and foreign derived demands, each of which satisfies a recursion of the same form (with foreign prices $\tau_j P_{i,n}$ moving one-for-one with $P_{i,n}$); the elasticity of a sum is the expenditure-share-weighted average of the component elasticities, which gives the stated convex combination. (iii) The right-hand side of (A.1) is a convex combination of 1 and $s_{i,n+1} \zeta_{i,n+1}$ with weights $\lambda_{i,n+1}$ and $1 - \lambda_{i,n+1}$. Because $s_{i,n+1} > 0$ and, by backward induction from (i), $\zeta_{i,n+1} > 0$, the second term is nonnegative and $\zeta_{i,n} \geq \lambda_{i,n+1} > 0$. The same argument applied to the foreign component yields the bound $\min\{\lambda_{i,n+1}, \lambda_{j,n+1}\}$ in the exported case. Together with (i) this gives $\zeta_{i,n} > 0$ for all active stages. $\square$

Lemma 7 (Sufficient conditions for a downward-sloping labor demand). *Fix $d \in \mathcal{D}$ and let $\varepsilon_{i,n} \in (0, 1]$ solve (36). Because labor is paid out of variable cost, $\omega_i l_{i,n}^* = \lambda_{i,n} V_{i,n} X_{i,n}$*

by (16), write

$$\theta_{i,n} \equiv \frac{\lambda_{i,n} V_{i,n} X_{i,n}}{\omega_i E_i^d}, \quad \sum_{n \leq \bar{n}_i} \theta_{i,n} = 1, \quad (\text{A.2})$$

for the share of stage n in the domestic wage bill, and let

$$\varepsilon_{i,n}^V \equiv \frac{\partial \ln V_{i,n}}{\partial \ln \omega_i} = \lambda_{i,n} + (1 - \lambda_{i,n}) \varepsilon_{i,n-1} \mathbf{1}\{n-1 \notin \mathcal{K}_i\} \in (0, 1], \quad \varepsilon_{i,n} = s_{i,n} \varepsilon_{i,n}^V, \quad (\text{A.3})$$

denote the wage elasticity of variable cost, where $s_{i,n} = V_{i,n}/P_{i,n}$ is the variable-cost share defined in (35); the second equality is Lemma 8. Then

$$\eta_i = \underbrace{\sum_{n \leq \bar{n}_i} \theta_{i,n} (1 - \varepsilon_{i,n}^V)}_{\text{cost-share (substitution) term}} + \underbrace{\sum_{n \leq \bar{n}_i} \theta_{i,n} \varepsilon_{i,n} \zeta_{i,n}}_{\text{scale term}}, \quad \zeta_{i,n} \equiv -\frac{\partial \ln X_{i,n}}{\partial \ln P_{i,n}} > 0, \quad (\text{A.4})$$

where $\zeta_{i,n} > 0$ for every active stage by Lemma 6(iii), with the explicit recursion (A.1). Hence $\eta_i > 0$, and $E_i^d(0^+) = +\infty$, $E_i^d(\infty) = 0$ follow from $\lim_{\omega_i \rightarrow 0^+} P_{i,n} = 0$ and $\lim_{\omega_i \rightarrow \infty} P_{i,n} = \infty$ in Lemma 3.

Proof. Differentiate the labor-market condition (32) stage by stage. Stage- n labor use is $\lambda_{i,n} V_{i,n} X_{i,n} / \omega_i$ by (16), so

$$\frac{\partial \ln(\lambda_{i,n} V_{i,n} X_{i,n} / \omega_i)}{\partial \ln \omega_i} = \varepsilon_{i,n}^V - 1 - \varepsilon_{i,n} \zeta_{i,n},$$

using (A.3) for the first term, Lemma 3 for $\partial \ln P_{i,n} / \partial \ln \omega_i = \varepsilon_{i,n}$, and the definition of $\zeta_{i,n}$. Aggregating with the weights (A.2) gives (A.4). Since $\varepsilon_{i,n}^V \leq 1$, the first term is nonnegative and strictly positive whenever some stage imports its input ($n-1 \in \mathcal{K}_i$, so $\varepsilon_{i,n}^V = \lambda_{i,n} < 1$). The scale term is strictly positive by Lemma 6. Hence $\eta_i > 0$. $\square$

By Assumption 4, which Lemma 7 shows to be implied by (36) and (53), aggregate labor demand $E_h^d(\omega_h)$, defined as the left-hand side of (32), is strictly decreasing, with

$$\eta_h \equiv -\frac{\partial \ln E_h^d}{\partial \ln \omega_h} > 0, \quad d \ln E_h^d = -\eta_h d \ln \omega_h. \quad (\text{A.5})$$

The participation rate enters allocations only through the employment identity $E_h = H_h N_h L_h$ and (32); it does not shift the wage-setting curve (8).

Wage response to participation. Combining $\omega_h = a_h H_h^{\beta_h}$ from (8) with $E_h = H_h N_h L_h$ and (A.5) gives

$$\kappa_\omega \equiv \frac{d \ln \omega_h}{d \ln N_h} = - \frac{\beta_h}{1 + \beta_h \eta_h} < 0, \quad (\text{A.6})$$

well defined whenever $1 + \beta_h \eta_h > 0$, which holds for all $\beta_h, \eta_h > 0$. For a general wage curve $\omega_h = W_h(H_h)$ with $W_h' > 0$, (A.6) holds with β_h replaced by the local elasticity $\beta_h(H_h) \equiv W_h'(H_h) H_h / W_h(H_h) > 0$, so no result below depends on the iso-elastic form.

Price elasticities. Let $\varepsilon_{h,n} \equiv \partial \ln P_{h,n} / \partial \ln \omega_h$ denote the stage-level pass-through of Lemma 3, which satisfies the recursion (36), and write

$$\varepsilon_h^t \equiv \varepsilon_{h,\bar{n}_h} = \frac{\partial \ln P_{h,\bar{n}_h}}{\partial \ln \omega_h} \in (0, 1]. \quad (\text{A.7})$$

Intensive-margin sourcing responses are governed by (53) with $\sigma_{Xh} = 1 + \alpha_h > 1$, and the extensive margin by (12)–(13).

Lemma 8 (Stage-level pass-through). *Under Assumption 2, the elasticities $\varepsilon_{i,n} \equiv \partial \ln P_{i,n} / \partial \ln \omega_i$ satisfy (36) for every $i \in \{h, f\}$ and $n \leq \bar{n}_i$, where $s_{i,n} \in (0, 1]$ is the variable-cost share defined in (35). Hence $\varepsilon_{i,n} \in (0, 1]$. Moreover $\varepsilon_{i,n} < 1$ for every $n \geq 2$ if $\chi > 0$; if $\chi = 0$, $\varepsilon_{i,n} < 1$ if and only if some upstream stage $m < n$ is imported.*

Proof. By Assumption 2 the primary stage uses labor only and bears no coordination charge, so $P_{i,1} = \omega_i$, $s_{i,1} = 1$ and $\varepsilon_{i,1} = 1$. For $n \geq 2$, write $P_{i,n} = V_{i,n} + \chi(n-1)$ with

$$V_{i,n} = \frac{1}{A_{i,n}} \left(\frac{\omega_i}{\lambda_{i,n}} \right)^{\lambda_{i,n}} \left(\frac{\hat{P}_{i,n-1}}{1 - \lambda_{i,n}} \right)^{1 - \lambda_{i,n}}, \quad \hat{P}_{i,n-1} = \min\{P_{i,n-1}, \tau_i P_{j,n-1}\}, \quad (\text{A.8})$$

as in (12). Since $\chi(n-1)$ does not depend on ω_i ,

$$\varepsilon_{i,n} = \frac{\partial \ln P_{i,n}}{\partial \ln \omega_i} = \frac{V_{i,n}}{P_{i,n}} \frac{\partial \ln V_{i,n}}{\partial \ln \omega_i} = s_{i,n} \frac{\partial \ln V_{i,n}}{\partial \ln \omega_i}.$$

Log-differentiating (A.8) gives $\partial \ln V_{i,n} / \partial \ln \omega_i = \lambda_{i,n} + (1 - \lambda_{i,n}) \partial \ln \hat{P}_{i,n-1} / \partial \ln \omega_i$, where $\partial \ln \hat{P}_{i,n-1} / \partial \ln \omega_i = \varepsilon_{i,n-1}$ if $n-1 \notin \mathcal{K}_i$ and $= 0$ if $n-1 \in \mathcal{K}_i$, because $\tau_i P_{j,n-1}$ is

invariant to ω_i . Combining the two displays yields (36). For the bounds, $\lambda_{i,n} \in (0, 1]$ and, by induction, $\varepsilon_{i,n-1} \in (0, 1]$ imply that the bracket in (36) lies in $(0, 1]$; multiplying by $s_{i,n} \in (0, 1]$ gives $\varepsilon_{i,n} \in (0, 1]$. If $\chi > 0$, then $s_{i,n} < 1$ for all $n \geq 2$ and hence $\varepsilon_{i,n} < 1$. If $\chi = 0$, then $s_{i,n} = 1$, and the bracket equals 1 exactly when $\varepsilon_{i,n-1} = 1$ and $n - 1 \notin \mathcal{K}_i$; iterating, $\varepsilon_{i,n} = 1$ if and only if no stage $m < n$ is imported. $\square$

Lemma 9 (Dominant diagonal of the labor-market system). *Fix $d \in \mathcal{D}$ and let $Z_i(\omega_h, \omega_f) \equiv E_i^d(\omega_h, \omega_f) - E_i^s(\omega_i)$ with $E_i^s(\omega_i) = H_i(\omega_i)N_iL_i$ and $H_i(\omega_i) = (\omega_i/a_i)^{1/\beta_i}$ from (8). The Jacobian $J \equiv \partial(Z_h, Z_f)/\partial(\ln \omega_h, \ln \omega_f)$ satisfies $J_{ii} < 0$, $J_{ij} \geq 0$ for $j \neq i$, and $|J_{ii}| > |J_{ij}|$; hence J is nonsingular.*

Proof. By (A.5), $\partial E_i^d/\partial \ln \omega_i = -\eta_i E_i^d < 0$, while $\partial E_i^s/\partial \ln \omega_i = E_i^s/\beta_i > 0$, so $J_{ii} < 0$. The foreign wage enters E_i^d only through $\tau_i P_{j,n-1}$ for $n - 1 \in \mathcal{K}_i$ in (12) and through the CES demands (24)–(31); a higher ω_j raises the cost of imported stages and, by (53), shifts expenditure toward domestic stages, so $J_{ij} \geq 0$. Finally, ω_i enters every stage with weight $\lambda_{i,n}$ and additionally enters E_i^s , whereas ω_j loads at most $(1 - \lambda_{i,n})s_{i,n} < 1$ on any stage and never enters E_i^s ; hence $|J_{ii}| > |J_{ij}|$, and J has a strictly dominant negative diagonal. This is the property invoked in Section 2.2.3 and in Assumption 5. $\square$

Lemma 10 (Robustness to relaxing the small-open-economy restriction). *Let J denote the labor-market Jacobian of Lemma 9, and drop Assumption 5. Then for any $\tau_i \geq 1$,*

$$\left| \frac{d \ln \omega_f}{d \ln N_h} \right| < \left| \frac{d \ln \omega_h}{d \ln N_h} \right| = |\kappa_\omega| + O(|J_{hf}J_{fh}|),$$

so the signs in Propositions 1–3 are unchanged.

Proof. Total differentiation of the two labor-market conditions (32) gives $J(d \ln \omega_h, d \ln \omega_f)' = (\beta_h^{-1} d \ln N_h, 0)'$ up to scale. Cramer's rule yields $d \ln \omega_f / d \ln \omega_h = -J_{fh}/J_{ff}$, and strict diagonal dominance of J (Lemma 9) gives $|J_{fh}| < |J_{ff}|$, hence the strict inequality. Diagonal dominance follows from $\varepsilon_{i,n} \in (0, 1]$ in (36) together with $\tau_i \geq 1$, which weakly attenuates every cross-country pass-through term relative to its own-country counterpart. $\square$

A.2 Proof of Lemma 3

Proof. Fix $(\bar{n}_i, \mathcal{K}_i)$, which is legitimate for $\omega_i \notin \Omega_i^*$ by Lemma 14. We show $\partial P_{i,n}/\partial \omega_i > 0$ for all $n \leq \bar{n}_i$ by induction. *Base step.* By Assumption 2 the primary stage uses labor only, so $P_{i,1} = \omega_i$ and $\partial P_{i,1}/\partial \omega_i = 1 > 0$. This follows from Assumption 2 and is not a normalization. *Induction step.* Suppose $\partial P_{i,n-1}/\partial \omega_i > 0$. If stage $n-1$ is sourced domestically, $\hat{P}_{i,n-1} = P_{i,n-1}$ by (12) and (A.8) gives

$$\frac{\partial P_{i,n}}{\partial \omega_i} = V_{i,n} \left[\frac{\lambda_{i,n}}{\omega_i} + \frac{1 - \lambda_{i,n}}{P_{i,n-1}} \frac{\partial P_{i,n-1}}{\partial \omega_i} \right] > 0. \quad (\text{A.9})$$

If it is imported, $\hat{P}_{i,n-1} = \tau_i P_{j,n-1}$ is independent of ω_i and

$$\frac{\partial P_{i,n}}{\partial \omega_i} = V_{i,n} \frac{\lambda_{i,n}}{\omega_i} > 0. \quad (\text{A.10})$$

The induction is complete. Since $\chi(n-1)$ is additive and wage invariant, it drops out of (A.9)–(A.10), and the elasticities implied by these expressions are exactly (36). $\square$

A.3 The chain-length margin

Recall from (18) the price change from adding one stage,

$$\Delta_i(n; \omega_i, \tau_i) \equiv P_{i,n+1} - P_{i,n} = [V_{i,n+1} - V_{i,n}] + \chi, \quad (\text{A.11})$$

so that (17), equivalently (19) under Assumption 3, is solved by the smallest $\bar{n}_i$ with $\Delta_i(\bar{n}_i) \geq 0$.

Lemma 11 (Existence of an equilibrium configuration). *The best-response map $\mathcal{B} : \mathcal{D} \rightrightarrows \mathcal{D}$ defined in (37) has a fixed point.*

Proof. By Section A.3 and Assumption 3, $\bar{n}_i(\omega_i)$ is a nondecreasing step function of ω_i and $\mathcal{K}_i(\omega_i)$ is nonincreasing in the set-inclusion order; hence $d \mapsto d'$ is monotone in the product order on $\mathcal{D}$ once $\omega^*(d)$ is shown to be monotone in d , which follows from strict monotonicity of $Z_i(\cdot; d)$ and Lemma 3. Since $\mathcal{D}$ is a finite lattice and $\mathcal{B}$ is a monotone self-map with nonempty values, Tarski's fixed-point theorem applies and the set of fixed points is a nonempty complete lattice. $\square$

Monotone comparative statics of the chain-length margin

Throughout this subsection we work with the main-text sign convention $\Delta_i(n; \omega_i, \tau_i) = P_{i,n+1} - P_{i,n} = [V_{i,n+1} - V_{i,n}] + \chi$ of (18), which is related to (A.11) by $\Delta_{i,n} = -\Delta_i(n-1)$; all statements below translate verbatim after this change of sign. Let the objective in (17) be $F_i(n; \omega_i) \equiv P_{i,n}(\omega_i, \tau_i)$, defined on the integer lattice $\mathbb{N}^+$ with parameter $\omega_i \in \mathbb{R}_{++}$, and let

$$\mathcal{N}_i(\omega_i) \equiv \arg \min_{n \in \mathbb{N}^+} F_i(n; \omega_i), \quad \bar{n}_i(\omega_i) \equiv \min \mathcal{N}_i(\omega_i).$$

Lemma 12 (Increasing differences of the stage margin). *Along a fully domestically sourced chain ($\mathcal{K}_i = \emptyset$), $V_{i,n} = \omega_i v_{i,n}$ with $v_{i,n}$ independent of ω_i and strictly decreasing in n (Lemma 1). Hence*

$$\Delta_i(n; \omega_i) = -\omega_i(v_{i,n} - v_{i,n+1}) + \chi, \quad \frac{\partial \Delta_i(n; \omega_i)}{\partial \omega_i} = -(v_{i,n} - v_{i,n+1}) < 0, \quad (\text{A.12})$$

so that $-F_i(n; \omega_i)$ has strictly increasing differences in $(n; \omega_i)$: for $n' > n$ and $\omega'_i > \omega_i$,

$$[F_i(n'; \omega'_i) - F_i(n; \omega'_i)] - [F_i(n'; \omega_i) - F_i(n; \omega_i)] = -(\omega'_i - \omega_i) \sum_{m=n}^{n'-1} (v_{i,m} - v_{i,m+1}) < 0.$$

If some stages are imported, $\mathcal{K}_i \neq \emptyset$, the same inequality holds with $v_{i,n}$ replaced by the wage-elastic component of $V_{i,n}$ implied by (A.9)–(A.10), because imported-input prices $\tau_i P_{j,n}$ are independent of ω_i and the overhead $\chi(n-1)$ is wage invariant.

Proof. Immediate from $V_{i,n} = \omega_i v_{i,n}$, homogeneity of degree one of $V_{i,n}$ in ω_i under domestic sourcing, $v_{i,n} - v_{i,n+1} > 0$ by Lemma 1, and the telescoping of $F_i(n'; \cdot) - F_i(n; \cdot) = \sum_{m=n}^{n'-1} \Delta_i(m; \cdot)$. The wage-invariant term $\chi(n-1)$ cancels in the double difference. $\square$

Lemma 13 (Topkis' theorem on the integer lattice). *Let $F_i(\cdot; \omega_i)$ satisfy Assumption 3 (strict single crossing of $\Delta_i(n; \omega_i)$ in n) together with the increasing-differences property of Lemma 12. Then:*

- (i) *For every $\omega_i > 0$ the minimizer set $\mathcal{N}_i(\omega_i)$ is nonempty, finite, and characterized by the threshold rule (19): $\bar{n}_i$ is the smallest n with $\Delta_i(n) \geq 0$.*

(ii) $\mathcal{N}_i(\cdot)$ is increasing in ω_i in the strong set order; in particular $\bar{n}_i(\omega_i)$ is nondecreasing in ω_i , which is (A.13).

(iii) $\bar{n}_i(\cdot)$ is a nondecreasing step function: there is a finite set $\Omega_i^* \subset \mathbb{R}_{++}$ on whose complement $\bar{n}_i$ is locally constant, and $\bar{n}_i$ jumps only upward, by construction at points of Ω_i^* .

Proof. (i) *Existence and the threshold rule.* Since $V_{i,n} \geq 0$ while $\chi(n-1) \rightarrow \infty$ as $n \rightarrow \infty$, $F_i(n; \omega_i) \rightarrow \infty$, so the minimum is attained on a finite subset of $\mathbb{N}^+$ and $\mathcal{N}_i(\omega_i) \neq \emptyset$. Necessity of (19) is immediate from optimality of $\bar{n}_i$ against $\bar{n}_i \pm 1$. For sufficiency, let n^* be the crossing point of Assumption 3 and let $\bar{n}_i$ be the smallest n with $\Delta_i(n) \geq 0$. For $n > \bar{n}_i$, $F_i(n) - F_i(\bar{n}_i) = \sum_{m=\bar{n}_i}^{n-1} \Delta_i(m) \geq 0$ because single crossing implies $\Delta_i(m) \geq 0$ for all $m \geq \bar{n}_i$; for $n < \bar{n}_i$, $F_i(\bar{n}_i) - F_i(n) = \sum_{m=n}^{\bar{n}_i-1} \Delta_i(m) < 0$ because $\Delta_i(m) < 0$ for all $m < \bar{n}_i$. Hence $\bar{n}_i$ is a global minimizer, and strictness of the crossing makes it unique. (ii) *Monotonicity.* Take $\omega'_i > \omega_i$, $n \in \mathcal{N}_i(\omega_i)$ and $n' \in \mathcal{N}_i(\omega'_i)$, and suppose, contrary to the strong set order, that $n' < n$. Optimality of n at ω_i gives $F_i(n; \omega_i) \leq F_i(n'; \omega_i)$, i.e. $F_i(n; \omega_i) - F_i(n'; \omega_i) \leq 0$. By Lemma 12 applied to the pair $(n' < n)$ and $(\omega_i < \omega'_i)$,

$$F_i(n; \omega'_i) - F_i(n'; \omega'_i) < F_i(n; \omega_i) - F_i(n'; \omega_i) \leq 0,$$

so $F_i(n; \omega'_i) < F_i(n'; \omega'_i)$, contradicting optimality of n' at ω'_i . Therefore $n' \geq n$, i.e. $\min \mathcal{N}_i(\cdot)$ and $\max \mathcal{N}_i(\cdot)$ are both nondecreasing, which is exactly Topkis' conclusion and yields (A.13). Equivalently, in threshold form: by (A.12), $\partial \Delta_i(n; \omega_i) / \partial \omega_i < 0$ shifts the function $n \mapsto \Delta_i(n)$ pointwise *downward* when ω_i rises, so the smallest n at which it turns nonnegative weakly rises. (iii) *Step structure.* $\Delta_i(n; \cdot)$ is continuous and, by (A.12), strictly decreasing in ω_i for each n ; hence for each $n \leq \bar{n}_i^{\max}$ the equation $\Delta_i(n; \omega_i) = 0$ has at most one solution $\omega_i^{(n)}$. Because $\chi(n-1) \rightarrow \infty$ caps the relevant range of n at a finite $\bar{n}_i^{\max}$ uniformly on compacta of ω_i , the set $\Omega_i^* \equiv \{\omega_i^{(n)}\}_{n \leq \bar{n}_i^{\max}}$ is finite. Off Ω_i^* all inequalities in (19) are strict, so $\bar{n}_i$, and with it the sourcing set $\mathcal{K}_i$ selected at $\bar{n}_i$, is locally

constant; at points of Ω_i^* it jumps by a positive integer. This proves Lemma 14. $\square$

On the integer lattice quasi-convexity of $n \mapsto F_i(n; \omega_i)$ restricts each cross-section but is not preserved under the parameter shift: it permits a minimizer set that is a non-order-convex collection of plateaus, along which the selection $\bar{n}_i(\omega_i)$ may fall as ω_i rises. The proof of part (ii) uses only the double-difference inequality of Lemma 12, i.e. a joint property of $(n; \omega_i)$, which quasi-convexity does not deliver. Single crossing in n is what converts (17) into the threshold rule (19), and increasing differences in $(n; \omega_i)$ is what makes that threshold monotone. It is the conjunction of the two, not either alone, that yields (A.13) and hence $d\bar{n}_h/dN_h \leq 0$ in Proposition 3 once combined with $\kappa_\omega < 0$ from Lemma 5.

If $\lambda_{i,n} = \lambda_i$ for all n and $A_{i,n} = n^c$ with $c \in (0, 1)$, then $\ln A_{i,n}$ is concave in n , $v_{i,n} - v_{i,n+1}$ is positive and decreasing in n , and $\Delta_i(n; \omega_i) = -\omega_i(v_{i,n} - v_{i,n+1}) + \chi$ is strictly increasing in n ; strict single crossing then holds automatically. With stage-varying $\lambda_{i,n}$, $n \mapsto v_{i,n} - v_{i,n+1}$ need not be monotone and multiple local minima can arise; in that case $\bar{n}_i$ is still well defined as the global minimizer in (17) by part (i) of Lemma 13's existence argument, but part (ii) may fail, and $\bar{n}_i$ need not be monotone in ω_i .

Lemma 14 (Threshold structure). *For each i there is a finite set $\Omega_i^* \subset \mathbb{R}_{++}$ such that, for $\omega_i \notin \Omega_i^*$, the pair $(\bar{n}_i, \mathcal{K}_i)$ is locally constant in ω_i , and*

$$\frac{\partial \bar{n}_i}{\partial \omega_i} \geq 0, \quad (\text{A.13})$$

with jumps only at $\omega_i \in \Omega_i^$.*

Proof. Because $\chi(n-1)$ diverges in n while stage efficiency gains are bounded by Assumption 2, $\Delta_{i,n} < 0$ for all n above a finite bound, so $\bar{n}_i$ is finite and $\mathcal{D}$ is finite, as in Section 2.2.3(ii). Under Assumption 3 the variable-cost saving is proportional to the wage, so $\Delta_{i,n}(\omega_i) = \omega_i v_{i,n} + \chi \zeta_{i,n}$ with $v_{i,n} < 0$ and $\zeta_{i,n} > 0$; hence $\partial \Delta_{i,n} / \partial \omega_i = v_{i,n} < 0$ and $\Delta_{i,n}$ crosses zero exactly once, at $\omega_i^{(n)} = -\chi \zeta_{i,n} / v_{i,n}$. Collect these roots in Ω_i^* . Off

Ω_i^* the set $\{n : \Delta_{i,n} \leq 0\}$ is nondecreasing in ω_i in the set-inclusion order, which is (A.13), and by continuity of stage prices (Lemma 3) the comparison $P_{i,n}$ versus $\tau_i P_{j,n}$ in (13) is locally constant as well. $\square$

Lemma 15 (The critical set is negligible). *The set Θ^c of parameters at which $\Delta_{i,n} = 0$ or $\tau_i P_{j,n-1} - P_{i,n-1} = 0$ for some (i, n) is a finite union of codimension-one C^1 hypersurfaces; it has empty interior and Lebesgue measure zero.*

Proof. Each defining function is C^1 by (A.8) with nonvanishing gradient: $\partial \Delta_{i,n} / \partial \omega_i = v_{i,n} < 0$ by Assumption 3, and $\partial(\tau_i P_{j,n-1} - P_{i,n-1}) / \partial \tau_i = P_{j,n-1} > 0$ by Lemma 3. The implicit-function theorem makes each zero set a codimension-one hypersurface, and $\mathcal{D}$ finite implies finitely many such surfaces. This establishes the claim in Section 2.2.3(iii). $\square$

A.4 Wage and price effects of participation

Lemma 16. *Under Assumption 5, a rise in N_h lowers ω_h and every domestic stage price, and raises the coordination-to-wage ratio:*

$$\frac{d \ln P_{h,n}}{d \ln N_h} = \varepsilon_{h,n} \kappa_\omega < 0 \quad \text{for all } n \leq \bar{n}_h, \quad \frac{d \ln(\chi / \omega_h)}{d \ln N_h} = -\kappa_\omega > 0. \quad (\text{A.14})$$

Proof. A rise in N_h enlarges the participating pool $N_h L_h$, so (32) requires a lower employment rate and (8) a lower wage; formally $\kappa_\omega < 0$ by (A.6). Applying Lemma 3 and Lemma 8 gives the first part of (A.14), while χ is wage invariant, giving the second. Under Assumption 5 the imported alternatives $\tau_h P_{f,n}$ are unchanged, so $P_{h,n} / (\tau_h P_{f,n})$ falls at every stage. $\square$

A.5 Proof of Proposition 1

Lemma 17 (Net sign of the share-based GVC indicators). *Fix $d \in \mathcal{D}$ and evaluate at $\Theta \setminus \Theta^c$ under Assumption 5. Let*

$$B_h = \frac{\sum_{n \leq \bar{n}_h} \tau_h P_{f,n-1} M_{h,n}}{\sum_{n \leq \bar{n}_h} \hat{P}_{h,n-1} X_{h,n}}, \quad A_h = \frac{\sum_{n \leq \bar{n}_h} P_{h,n} X_{hf,n}}{\sum_{n \leq \bar{n}_h} P_{h,n} X_{h,n}}.$$

Then

$$\frac{d \ln B_h}{d \ln N_h} = -(\sigma_{Xh} - 1) \left(1 - \bar{s}_h\right) \bar{\varepsilon}_h \kappa_\omega < 0, \quad \frac{d \ln A_h}{d \ln N_h} = -(\sigma - 1) (1 - A_h) \varepsilon_h^t \kappa_\omega > 0, \quad (\text{A.15})$$

where $\bar{s}_h$ and $\bar{\varepsilon}_h$ are expenditure-weighted averages of $s_{h,n}$ and $\varepsilon_{h,n}$, and $\varepsilon_h^t \equiv \varepsilon_{h,\bar{n}_h}$. In particular, the quantity (scale) terms in the numerator and the denominator are common and cancel, so the net signs are governed by the relative-price terms alone.

Proof. Both A_h and B_h are ratios of expenditures measured in the same units, so a common proportional change in all stage outputs $\{X_{h,n}\}_n$ leaves them unchanged. Writing each share in CES form, $1 - s_{h,n} = (\tau_h P_{f,n-1})^{1-\sigma_{Xh}} / \hat{P}_{h,n-1}^{1-\sigma_{Xh}}$, and differentiating with the help of (53) gives $d \ln(1 - s_{h,n}) / d \ln \omega_h = (\sigma_{Xh} - 1) s_{h,n} \varepsilon_{h,n-1}$. Aggregating over stages with expenditure weights and applying $d \ln \omega_h / d \ln N_h = \kappa_\omega < 0$ from (A.6) yields the first expression in (A.15). The same argument applied to the final-stage export share, using the CES final-goods demand system in (6), yields the second. Under Assumption 5 the foreign prices $\{P_{f,n}\}_n$ and ω_f are fixed, so no offsetting foreign-price movement arises. $\square$

Proof. Step 1 (backward participation). By (A.14) and (53) with $\sigma_{Xh} > 1$, the fall in domestic stage prices lowers the imported expenditure share on the intensive margin. On the extensive margin, since $P_{h,n} / (\tau_h P_{f,n})$ falls at every stage, the sourcing criterion (13) is satisfied for a weakly smaller set of stages, so $\mathcal{K}_h$ weakly contracts. Both margins lower imported intermediate expenditure relative to gross output, so backward participation falls,

$$\frac{\partial \text{GVC}_{b,h}}{\partial N_h} < 0. \quad (\text{A.16})$$

Step 2 (forward participation). Under Assumption 5, foreign prices are fixed while $P_{h,\bar{n}_h}$ falls by (A.7) and (A.14). By the foreign counterparts of (24) and (53) with $\sigma_{Xf} = 1 + \alpha_f > 1$, foreign absorption of domestic stages rises, and by the foreign counterpart of

(13) the set $\mathcal{K}_f$ weakly expands toward home stages. Hence

$$\frac{\partial \text{GVC}_{f,h}}{\partial N_h} > 0. \quad (\text{A.17})$$

Equations (A.16)–(A.17) are the claims of Proposition 1. $\square$

A.6 Proof of Proposition 2

Proof. The position index is $\text{GVC}_{\text{pos},h} = \ln(1 + A_h) - \ln(1 + B_h)$ with A_h and B_h defined in (50)–(51). Define

$$\kappa_A \equiv \frac{\partial \ln A_h}{\partial \ln \omega_h} < 0, \quad \kappa_B \equiv \frac{\partial \ln B_h}{\partial \ln \omega_h} > 0, \quad (\text{A.18})$$

whose signs are established in Steps 1–3 of Appendix A.5. For parameters outside Θ^c the configuration is locally constant by Lemma 14, so

$$\frac{d \text{GVC}_{\text{pos},h}}{d \ln N_h} = \kappa_\omega \left[\frac{A_h}{1 + A_h} \kappa_A - \frac{B_h}{1 + B_h} \kappa_B \right] > 0, \quad (\text{A.19})$$

because the bracket is strictly negative and $\kappa_\omega < 0$. $\square$

A.7 Proof of the chain-length proposition

Proof. By (A.14) the ratio χ/ω_h rises, so $\Delta_{h,n}$ in (A.11) falls for every n . Combining (A.13) with $\kappa_\omega < 0$ from (A.6), for $N'_h > N_h$ we have $\omega_h(N'_h) < \omega_h(N_h)$ and therefore

$$\bar{n}_h(N'_h) \leq \bar{n}_h(N_h), \quad \text{that is} \quad \frac{\partial \bar{n}_h}{\partial N_h} \leq 0, \quad (\text{A.20})$$

a step function with strict declines only at Ω_h^* . Since observed average production length is a strictly increasing transformation G of $\bar{n}_h$ with $G' > 0$, $\partial \text{PLY}_h / \partial N_h = G'(\bar{n}_h) \partial \bar{n}_h / \partial N_h \leq 0$. $\square$

A.8 Proof of Proposition 4

Proof. At fixed $\bar{n}_h$, Step 1 of Appendix A.5 shows $\mathcal{K}_h(N'_h) \subseteq \mathcal{K}_h(N_h)$ for $N'_h > N_h$, so the number of domestically performed stages $\bar{n}_h - |\mathcal{K}_h|$ weakly rises, while (A.20) shows $\bar{n}_h$ weakly falls. Hence the domestic share $1 - |\mathcal{K}_h|/\bar{n}_h$ weakly rises and the cross-border share $|\mathcal{K}_h|/\bar{n}_h$ weakly falls, with $|\mathcal{K}_h|$ itself weakly falling by (13) together with Lemma 16. $\square$

A.9 Trade-cost amplification

The trade-cost corollaries require that a higher τ_h strengthen each margin. We state the two conditions used and prove the cross-partials.

$$\frac{\partial^2 \ln(P_{h,n}/\tau_h P_{f,n})}{\partial \ln \omega_h \partial \ln \tau_h} \leq 0 \quad (\text{Condition AMP-S, sourcing margin}), \quad (\text{A.21})$$

$$\frac{\partial^2 \Delta_{h,\bar{n}_h+1}}{\partial \ln \omega_h \partial \ln \tau_h} \geq 0 \quad (\text{Condition AMP-L, chain-length margin}). \quad (\text{A.22})$$

Proof. Sourcing margin. A higher τ_h raises $\tau_h P_{f,n}/P_{h,n}$ at every stage. Under (A.21) a given proportional wage decline induces a weakly larger contraction of $\mathcal{K}_h$, so $\partial \kappa_B / \partial \ln \tau_h > 0$ and $\partial \kappa_A / \partial \ln \tau_h < 0$ in (A.18). Hence

$$\frac{\partial^2 \text{GVC}_{b,h}}{\partial N_h \partial \ln \tau_h} \leq 0, \quad \frac{\partial^2 \text{GVC}_{f,h}}{\partial N_h \partial \ln \tau_h} \geq 0, \quad \frac{\partial^2 \text{GVC}_{\text{pos},h}}{\partial N_h \partial \ln \tau_h} > 0, \quad (\text{A.23})$$

where the last inequality follows by differentiating (A.19) once more with respect to $\ln \tau_h$: the bracket becomes more negative while $\kappa_\omega < 0$. *Chain-length margin.* The threshold wage $\omega_h^{(\bar{n}_h)}(\tau_h)$ solves $\Delta_{h,\bar{n}_h+1}(\omega_h, \tau_h) = 0$, with $\partial \Delta_{h,\bar{n}_h+1} / \partial \omega_h < 0$ by Lemma 14. By the implicit-function theorem and (A.22), a higher τ_h steepens the wage sensitivity of this margin, so for $N'_h > N_h$ and $\tau'_h > \tau_h$,

$$\bar{n}_h(N'_h, \tau'_h) \leq \bar{n}_h(N'_h, \tau_h) \leq \bar{n}_h(N_h, \tau_h). \quad (\text{A.24})$$

Absent (A.22) the first inequality in (A.24) holds with equality and no chain-length amplification arises; amplification along this margin is therefore a conditional prediction. $\square$

A.10 Welfare decomposition

Proof. Household optimization subject to (3) with income (4) gives indirect utility $\omega_i E_i / P_{U_i}$. Using $E_i = H_i N_i L_i$ and dividing by L_i yields (38). Since Assumption 1 rules out any coordination endowment, labor income is total household income and no coordination-income term appears. Because firms earn zero profits and coordination services are the numeraire,

total household expenditure equals labor income, so no separate coordination-income adjustment is required. Define the income-adjustment term $\Lambda_h \equiv \omega_h H_h / p_h$ and

$$\eta_H \equiv -\frac{d \ln H_h}{d \ln \tau_h}, \quad \eta_p \equiv -\frac{d \ln p_h}{d \ln \tau_h}. \quad (\text{A.25})$$

Log-differentiating $\ln \Lambda_h = \ln \omega_h + \ln H_h - \ln p_h$ and substituting $d \ln \omega_h = \beta_h d \ln H_h$ from (8) gives

$$-\frac{d \ln \Lambda_h}{d \ln \tau_h} = (1 + \beta_h) \eta_H - \eta_p, \quad (\text{A.26})$$

so that $\Lambda_h > 1$ if and only if

$$(1 + \beta_h) \eta_H > \eta_p. \quad (\text{A.27})$$

The direction in which Λ_h departs from unity is therefore *not* signed by $\beta_h > 0$ alone. Two limits confirm this. If $\beta_h = 0$ and H_h is inelastic, then $\eta_H = 0$, (A.26) reduces to $-\eta_p$, and $\Lambda_h = 1$ whenever the domestic producer price is unchanged, recovering the standard sufficient-statistic formula. If η_p is large, the labor-market channel attenuates rather than amplifies the gains from trade.

Finally, totally differentiating the welfare measure and comparing with the Arkolakis–Costinot–Rodríguez-Clare formula, the wedge equals $(1 + \beta_h) d \ln H_h$, which is not a function of the domestic expenditure share and the trade elasticity alone. The model therefore lies outside that class whenever $\beta_h > 0$ and $d \ln H_h \neq 0$. $\square$